HARDWARE AND SYSTEMS

Hardware and low-level systems from first principles

UNIT 1

Circuits, Sensors and Signals

From electrons to measured, digitised and transmitted signals

1.1 Electricity, components and measurement

1.2 The analogue signal chain

1.3 Sensors with TI silicon

1.4 Power conversion and batteries

1.5 Signals on wires and RF fundamentals

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UNIT 1

About this unit

Unit 1 teaches a software engineer to measure circuits and signals. It starts with Ohm's law and a meter and ends with link budgets checked against an SDR. In between come op-amps, filters and converters, five TI sensors driven from their register maps, a buck converter taken apart with a scope, and the drone's battery characterised to the watt-hour. Every later unit depends on these measurements. The thrust stand, the radio link, the boards and the BMC all reuse what you build here.

Chapter	You build	Weeks
1.1	A measured divider, RC charge, MOSFET switch and a decoupling failure captured on the oscilloscope	2
1.2	Op-amp stages, a filter with a Bode plot, a characterised ADC, a load cell on a 24-bit converter	2
1.3	Library-free drivers and metrology for TI sensors; an error budget for the drone's current	2.5
1.4	Converter, protection and battery measurements; the drone and rack power trees	2.5
1.5	Reflections, S-parameters, a tuned antenna, an FM receiver, link budgets	2.5

Before you start

Work through the Handbook's setup checklist for the bench, the workstation and the Pi. The rest of the setup can wait for Unit 2.

Kit for this unit

The Handbook's kit part lists every item for this unit with its part number, seller and price. The totals, for items this unit needs first:

	Required	Optional
UK sellers, VAT included (ex-VAT prices × 1.2)	£3,212	£603
Overseas sellers, before import VAT and delivery	€145	US$26
Items with no checked price	1	1

Not priced: Isopropyl alcohol, 99 % or stronger, bought locally.

CHAPTER 1.1

Electricity, components and measurement

TIME	2 weeks
YOU BUILD	An LED circuit and a loaded divider explained by Thevenin; a fitted RC time constant; a characterised MOSFET switch with and without a flyback diode; a decoupling failure captured on the oscilloscope.
NEEDS	Bench kit (multimeter, oscilloscope, bench supply, breadboards, passives, discrete semiconductors, SN74HC04N and SN74HC14N, 12 V PWM fan).

WHAT THIS BUYS OMH

Every layer above this one depends on it. A rack module is a power tree (a 12 V or 48 V bus feeding dozens of point-of-load rails), a thermal budget (every watt in is a watt of heat out) and a large number of decoupling capacitors that keep rails quiet while PCIe lanes switch at gigahertz. A quadcopter has the same problems, with a battery instead of a bus and 60 A of motor current instead of a PSU. You will not design every converter yourself, but you will specify, review and debug them, and that work needs the intuition this chapter builds.

SKIP TEST

1. A 3.3 V rail with 10 µF of decoupling sees a 2 A load step lasting 1 µs. How far does the rail dip, ignoring ESR and inductance? What if there is also 5 nH of trace inductance between the capacitor and the load?
2. Why does an LED need a series resistor, while a resistor does not need a series LED?
3. Explain, in terms of charge, why an I^2C bus with a 10 kΩ pull-up cannot run at 400 kHz with 300 pF of bus capacitance.
4. What does a 10× oscilloscope probe actually do, and why does grounding it at the wrong place show ringing that is not there?
5. A MOSFET switches an inductive load off. Where does the stored energy go if there is no flyback diode?

If all five are easy, do Labs 1.1.3 and 1.1.4 only.

Core ideas

Voltage is energy per unit charge; current is charge per unit time. Power is their product, $P = VI$. In every hardware design, P is the heat you must remove and I is the current your connectors, traces and wires must carry.

Kirchhoff's laws hold for every circuit. The currents into a node sum to zero and the voltages around a loop sum to zero, because charge and energy are conserved. Ohm's law, $V = IR$, describes a material: resistors obey it, and diodes and transistors do not. Thevenin's theorem collapses any linear network seen from two terminals into one source V_{th} behind one resistance R_{th}. That resistance, R_{th}, tells you how stiff the node is, and every power and sensing design needs to know it.

A divider cannot supply power. Its output sags as soon as you draw current, in proportion to its Thevenin resistance (the two resistors in parallel). Sensing circuits use dividers, and power rails use regulators.

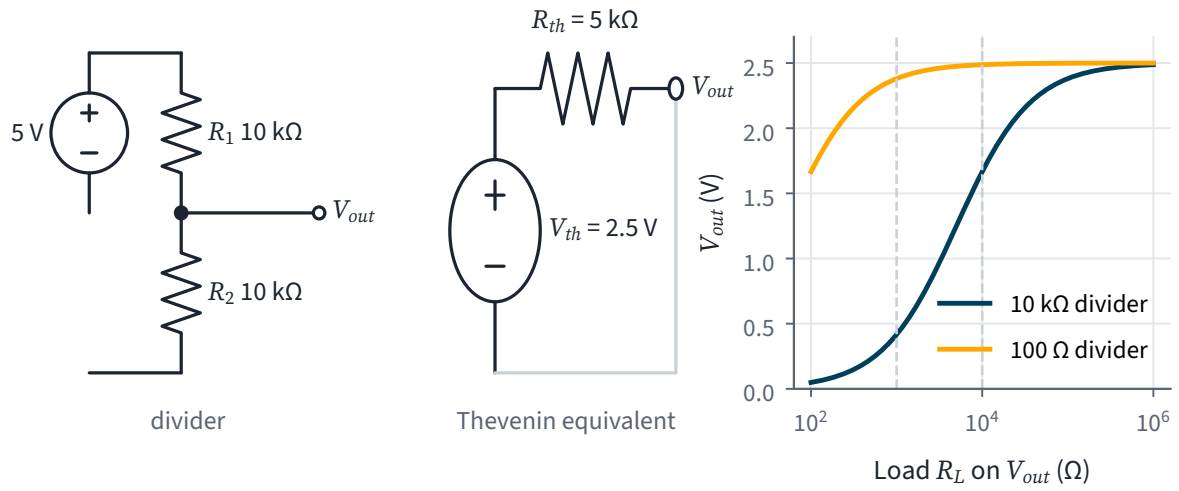


Figure 1.1.1. A divider and its Thevenin equivalent. Loading the output with R_L forms a second divider with R_{th}, which is why a 10 kΩ divider collapses under a 1 kΩ load.

Capacitors store charge and resist changes of voltage. $I = C\,\mathrm{d}V/\mathrm{d}t$. An RC network has the time constant $\tau = RC$; a step covers 63.2 % of its final value after one τ and 99.3 % after five. Decoupling works because a capacitor next to a load is a local reservoir that supplies fast current before the distant regulator can respond. Its usefulness at high frequency is limited by its own series inductance (ESL) and resistance (ESR), which is why boards use many small capacitors close to the pins rather than one large one far away.

Inductors store current and resist changes of current. $V = L\,\mathrm{d}I/\mathrm{d}t$. Every wire and trace has inductance, roughly 1 nH per millimetre for a wire or a trace far from any plane, and about a third of that for a trace over a plane. A fast current step through a long return path produces a voltage spike (ground bounce). Switching off an inductive load produces a spike large enough to destroy the switch unless something gives the current somewhere to go.

Diodes conduct exponentially. A small rise in forward voltage produces a large rise in current, so a diode or LED across a fixed voltage with no series resistor draws whatever the source will give. Typical forward voltages are 0.6 to 0.7 V for silicon, 0.2 to 0.4 V for Schottky and 1.8 to 3.2 V for LEDs, depending on colour. Power paths use Schottky diodes for their low drop and fast recovery.

A MOSFET is a voltage-controlled switch. Above the threshold $V_{\text{GS(th)}}$ a channel forms, and the on-resistance $R_{\text{DS(on)}}$ sets the conduction loss I^2R. The gate is a capacitor, so switching it takes charge and time, and during the transition the device is partly on and dissipates heavily. Switching loss scales with frequency, and conduction loss does not. A “logic-level” MOSFET is fully on at 3.3 or 5 V of gate drive, and many MOSFETs are not logic-level. Take $R_{\text{DS(on)}}$ from the curve of $R_{\text{DS(on)}}$ against V_{GS} at your gate drive, and do not rely on the headline number.

Open-drain outputs and pull-ups let many devices share a wire. Any device may pull the line low, and a resistor pulls it high when none does. The rise time is proportional to the pull-up resistance times the bus capacitance, and it limits the bus speed. I²C, SMBus, PMBus, interrupt and reset lines on every board you will build work this way.

Real sources have internal resistance. A USB port, a bench supply, a lithium-polymer pack and a server PSU are all Thevenin sources. The droop under a steady load and the dip under a step give their effective source impedance. A drone battery’s internal resistance sets how far its voltage sags when four motors spin up at once.

Measurement changes the circuit. A multimeter reports averages and cannot show anything fast. An oscilloscope probe adds 10 pF or more of capacitance, and its 15 cm ground lead is an inductor that rings with that capacitance near 100 MHz. A 10× probe divides the signal by ten and presents ten times the impedance of a 1× probe. Ground at the nearest ground to your signal with the shortest lead you have, using the spring tip for fast edges.

Dissipated power raises temperature. Every component has a thermal resistance from junction to ambient, θ_{JA} in kelvin per watt, and its temperature rise is $P\theta_{\text{JA}}$. A 2 W part with $\theta_{\text{JA}} = 60$ K/W runs 120 K

above ambient and fails. Copper pours, thermal vias, heatsinks and airflow reduce θ_{JA}. Rack modules and motor drivers are both designed around their thermal limits.

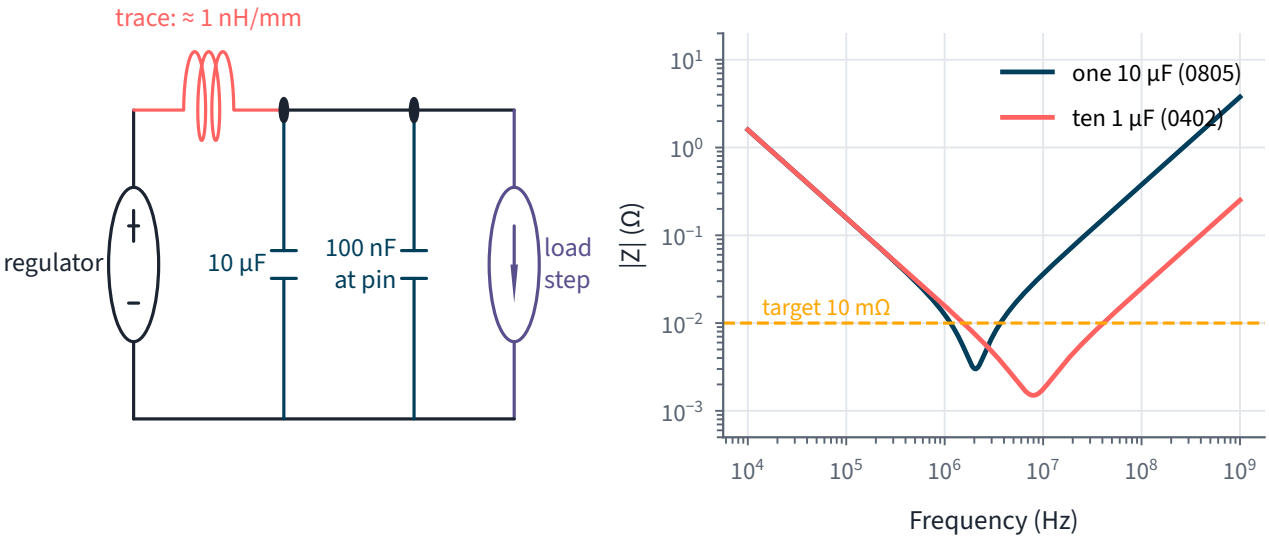


Figure 1.1.2. Why decoupling capacitors go next to the pins. Left: the loop a load-step current takes. Right: the impedance of one 10 µF ceramic and of ten 1 µF ceramics, each with realistic ESL and ESR. Above its self-resonant frequency a capacitor behaves as an inductor.

Reading

- Horowitz and Hill, *The Art of Electronics*, 3rd ed.: chapter 1 (all), sections 3.1 to 3.5 (FETs as switches).
- Scherz and Monk, *Practical Electronics for Inventors*, chapters 2 to 4, for a gentler first pass if AoE is too dense.
- One decoupling-placement application note from TI or Analog Devices (search either site for "decoupling capacitor placement"). Read it once now and again before your first layout.
- Falstad circuit simulator: reproduce every lab in simulation before building it, and keep the simulation link in your notebook.

Labs

LAB 1.1.1 Divider, LED and source resistance — 3 h

Goal. measure how a divider's output falls under load, and how stiff a real supply is.

1. Compute the series resistor for a red LED at 10 mA from 5 V, using the forward voltage from its datasheet. Build it on the breadboard from the bench supply. Measure the voltage across the LED and across the resistor, compute the actual current and compare it with your prediction.
2. Build a 2:1 divider from two 10 kΩ resistors on 5 V. Predict, then measure, the output unloaded, with a 10 kΩ load and with a 1 kΩ load.
3. Repeat with two 100 Ω resistors. Compute how much power the divider itself wastes in each case.
4. Measure the bench supply's output impedance: record its voltage at no load and at 500 mA into a power resistor, with the current limit set to 1 A. Repeat at a USB port with a USB load tester or a resistor on a breakout.

DONE WHEN

- ☐ Predicted and measured values agree within resistor tolerance (5 %) for every divider case.
- ☐ The 1 kΩ case is explained in the notebook with a drawn Thevenin equivalent and a one-line calculation.

- [ ] The bench supply's and the USB port's effective source resistances are computed from your two measurements each.

EVIDENCE FOR THE NOTEBOOK
- A table of predicted and measured values with the percentage error per row.
- Photographs of the breadboard and the meter for the LED case.

LAB 1.1.2 RC charging and a threshold crossing — 3 h

Goal. measure $\tau = RC$ and use the oscilloscope's single-shot trigger.

1. Build an RC network (10 kΩ, 10 µF) driven by a push button from 5 V, with a 100 kΩ discharge resistor across the capacitor. Capture the charge curve on the oscilloscope in single-shot mode, triggered on the rising edge.
2. Fit τ from the capture using the cursors at 63.2 % and by fitting the exponential to exported data in Python. Compare with RC and explain any discrepancy (electrolytic capacitors are often −20 %/ +80 %).
3. Predict the time to cross 2.0 V, a common logic-high threshold, and measure it.
4. Change R to 1 kΩ and repeat. Then capture the discharge curve and fit it.

DONE WHEN
- [ ] Your fitted τ is within 20 % of RC and the notebook explains the error with the capacitor's tolerance and the meter's measurement of C.
- [ ] The measured 2.0 V crossing time is within 10 % of your prediction from the fitted τ.

EVIDENCE FOR THE NOTEBOOK
- Scope captures of charge and discharge, the exported CSV and the Python fit.

LAB 1.1.3 The MOSFET as a switch — 4 h

Goal. measure conduction loss, switching loss and the turn-off spike of an inductive load.

1. Drive a 12 V fan from a 2N7000 and then from a TI CSD18534KCS (low-side switch, source to ground), with the gate driven by an NE555P astable at about 1 kHz. Measure V_{DS} while on and compute $R_{DS(on)}$ and the conduction loss for each transistor.
2. Add a 10 kΩ resistor in series with the gate. Capture gate and drain voltages together. Measure the transition time and estimate the switching energy as the area under $V_{DS} \times I_D$ during the transition.
3. Capture the drain at turn-off without a flyback diode, then with a 1N5819 across the fan. Measure the peak voltage in each case and explain it with $V = L\,\mathrm{d}I/\mathrm{d}t$.

SAFETY
The turn-off spike without a diode can exceed the 2N7000's 60 V rating. Keep the run short, use the avalanche-rated CSD18534KCS for this step and switch the supply off between changes.

DONE WHEN
- [ ] Your notebook says which transistor is logic-level at 5 V of gate drive and shows the $R_{DS(on)}$ curve from its datasheet that proves it.
- [ ] Two captures show the turn-off spike with and without the diode, with peak voltages labelled.
- [ ] Conduction and switching losses are computed for both transistors at 1 kHz and extrapolated to 100 kHz.

EVIDENCE FOR THE NOTEBOOK

- Scope captures with both channels labelled; the loss calculation.

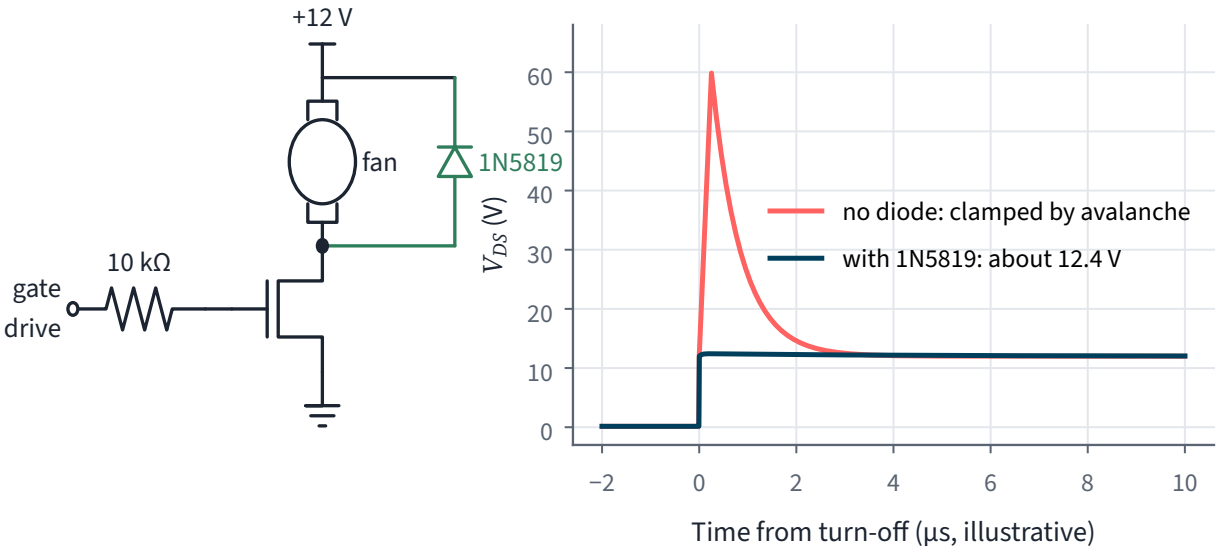


Figure 1.1.3. Low-side switch for an inductive load. Without the flyback diode the drain voltage at turn-off is set by $L\,\mathrm{d}I/\mathrm{d}t$. With the diode, the current circulates through it and decays.

LAB 1.1.4 Decoupling failure on the oscilloscope

3 h

Goal. measure how decoupling capacitors and their placement change the noise on a supply rail.

1. Build a fast square-wave source from an SN74HC14N (Schmitt inverter oscillator) driving a few hundred picofarads, powered through a pair of 50 cm breadboard wires from the bench supply. Fit no decoupling.
2. Probe the supply at the IC's power pin with the spring ground tip. Record the peak-to-peak noise.
3. Add a 100 nF ceramic directly across the IC's supply pins and record again. Then add a 10 µF electrolytic at the supply end of the long wires and record.
4. Move the 100 nF capacitor to the far end of the wires and record. Explain the difference in terms of the inductance between the capacitor and the pin.

DONE WHEN

- ☐ Three captures show the rail noise falling as decoupling is added, each with its peak-to-peak value.
- ☐ One capture shows that a capacitor 50 cm away barely helps, and the notebook estimates the inductance of the wires from the ringing frequency.

EVIDENCE FOR THE NOTEBOOK

- Four captures at the same vertical scale; your inductance estimate.

Problem set

1. Derive the RC step response from $I = C\,\mathrm{d}V/\mathrm{d}t$ and Ohm's law. Show where 63.2 % comes from.
2. A load steps by 3 A in 50 ns on a 1.0 V rail that may dip at most 30 mV. Ignoring inductance, how much decoupling capacitance is needed? Now add 2 nH between the capacitor and the load and compute the inductive dip alone. What does this tell you about where the capacitors go?
3. An I^2C bus has 250 pF of capacitance and needs a rise time (30 % to 70 %) under 300 ns for Fast-mode. What is the largest pull-up allowed? What is the smallest, given a 3 mA sink limit at 3.3 V?

4. A power MOSFET with $R_{DS(on)} = 25$ mΩ switches 10 A at 100 kHz with 50 ns transitions between 12 V and 0 V. Estimate conduction and switching loss. Which dominates, and what happens at 1 MHz?
5. A server's 12 V rail has 4,700 µF of bulk capacitance. A hot-swapped drive draws 4 A for 2 ms while its regulators start. How far does the rail dip if the PSU cannot respond in that time? Why do hot-swap controllers exist?
6. A 3S lithium-polymer pack reads 12.4 V at rest and 11.2 V while four motors draw 40 A. What is its internal resistance, how much power is lost inside the pack, and what does that do to the pack's temperature over a five-minute flight?
7. Explain why an SN74HC output driving a 1 m cable rings, and what a 33 Ω series resistor at the source does.
8. A USB port is specified at 5 V with up to 500 mA and up to 0.35 V of droop. What is its effective source resistance? What is the largest load step it can handle with a dip under 100 mV, given 10 µF at the load?

Deliverables and stretch

Deliverables. The notebook with predictions, measurements and captures for all four labs, and a one-page "power and decoupling" cheat sheet in your own words. You will reuse the cheat sheet in every board design review in Unit 3.

Stretch. Simulate Lab 1.1.4 in ngspice with a capacitor model that includes ESL and ESR, and plot impedance against frequency for one 10 µF capacitor against ten 1 µF capacitors. Build a current-limited hot-swap circuit with a P-channel MOSFET, a sense resistor and a comparator, and capture the controlled inrush.

COMPLETION CHECKLIST

Tick each line and date it. Do not start Chapter 1.2 until every line holds.

- ☐ **Lab 1.1.1:** divider table complete with errors under 5 %; Thevenin drawing for the 1 kΩ case; two source resistances computed.
- ☐ **Lab 1.1.2:** fitted τ within 20 % of RC for both resistor values, with the Python fit committed to the labs repository.
- ☐ **Lab 1.1.3:** logic-level verdict for both MOSFETs with the datasheet curve; turn-off captures with and without the diode.
- ☐ **Lab 1.1.4:** four rail-noise captures at one scale; the far-capacitor case explained with an inductance estimate.
- ☐ **Problem set:** all eight answered in writing with units, and each numeric answer checked by a second method or a simulation.
- ☐ **Cheat sheet** written and committed.
- ☐ **You can explain**, without notes: Thevenin equivalence, why $\tau = RC$ gives 63 %, why switching loss scales with frequency, and why a decoupling capacitor's position matters more than its value.

CHAPTER 1.2

The analogue signal chain

TIME	2 weeks
YOU BUILD	Two op-amp stages measured against their datasheets; a second-order filter with a measured Bode plot; a characterised 16-bit ADC with a demonstrated alias; a calibrated load cell on a 24-bit ADC that becomes Unit 3's thrust stand.
NEEDS	Bench kit, function generator, Raspberry Pi 5, LM358P and TLV2372IP op-amps, LM4040 reference, the Adafruit ADS1115 and Olimex BB-ADS1220 breakouts, the SparkFun 5 kg load cell.

WHAT THIS BUYS OMH

Every sensor in this course produces an analogue quantity that something must amplify, filter and digitise before software sees it: the shunt voltage behind a power reading, the bridge output of a load cell, the beat frequency of a radar. The least accurate stage limits the whole measurement, and that stage is almost never the one a datasheet's headline resolution describes. You will use this chapter to judge every ADC, front end and sensor you choose for OMH's boards and the drone.

SKIP TEST

1. An op-amp with a 1 MHz gain-bandwidth product is wired as a non-inverting amplifier with a gain of 21. What is its −3 dB bandwidth, and what limits a 4 V peak-to-peak output at 50 kHz if the slew rate is 0.5 V/µs?
2. Design a unity-gain Sallen-Key low-pass filter with a 1 kHz Butterworth response. Give component values and say what sets Q.
3. A 12-bit ADC samples at 10 kSa/s. Where does a 9.3 kHz tone appear in the sampled data, and what must sit in front of the ADC to stop it?
4. What is the thermal noise voltage of a 10 kΩ resistor over a 10 kHz bandwidth at room temperature, and how does it compare with one LSB of a 16-bit ADC with a ±2.048 V range?
5. Why does a ratiometric bridge measurement not care about the exact excitation voltage?

If all five are easy, do Labs 1.2.3 and 1.2.4 only.

Core ideas

The chain. A measurement passes through a transducer (physical quantity to voltage or current), conditioning (gain, level shift, impedance buffering), an anti-alias filter, a sample-and-hold, a quantiser, and digital filtering. Each stage adds error: offset, gain error, nonlinearity, noise, drift and bandwidth limits. You budget them the way you budget power.

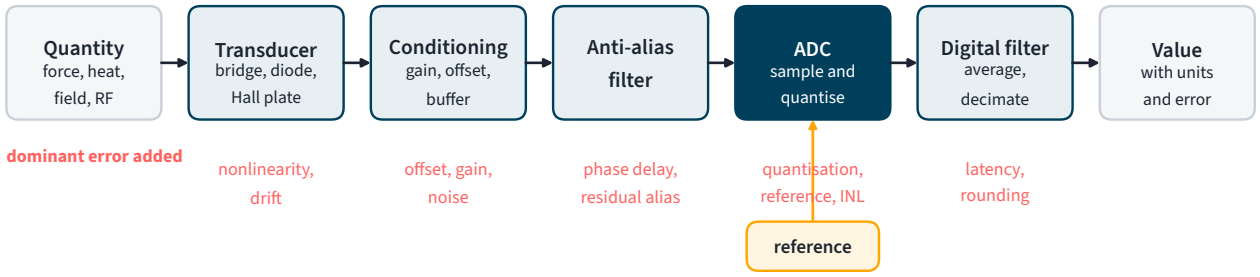


Figure 1.2.1. The signal chain, with the dominant error each stage adds. Unit 3's thrust stand, radar and power telemetry are all instances of this picture.

The op-amp, with negative feedback. An ideal op-amp has infinite gain, infinite input impedance and zero output impedance. With negative feedback two rules follow: the output moves until the two inputs are equal, and the inputs draw no current. The non-inverting amplifier has gain $G = 1 + R_f/R_g$, and the inverting amplifier has $G = -R_f/R_{\text{in}}$. A real op-amp departs from the ideal in ways the datasheet quantifies, and each departure becomes an error term:

- **Gain-bandwidth product.** Open-loop gain falls at 20 dB per decade, so the closed-loop bandwidth is roughly GBW divided by the noise gain $1 + R_f/R_g$. A 1 MHz part at a gain of 100 has about 10 kHz of bandwidth.
- **Slew rate.** The output can change only so fast. A sine of peak amplitude A at frequency f needs $2\pi f A$ volts per second; above that the output becomes a triangle.
- **Input offset voltage and bias current.** A few millivolts of offset, multiplied by the gain, appears at the output as a DC error. Zero-drift parts such as TI's OPA2333 reduce it to microvolts at the cost of bandwidth.
- **Supply rails.** "Rail-to-rail" inputs and outputs reach within tens of millivolts of the supplies. An LM358 on 5 V cannot drive its output above about 3.5 V, and a TLV2372 can reach nearly 5 V. On single-supply boards this determines whether your signal fits.
- **Stability.** A capacitive load or a feedback capacitor in the wrong place adds phase shift and turns an amplifier into an oscillator. Read the "capacitive load" figure in the datasheet before driving a cable or an ADC input.

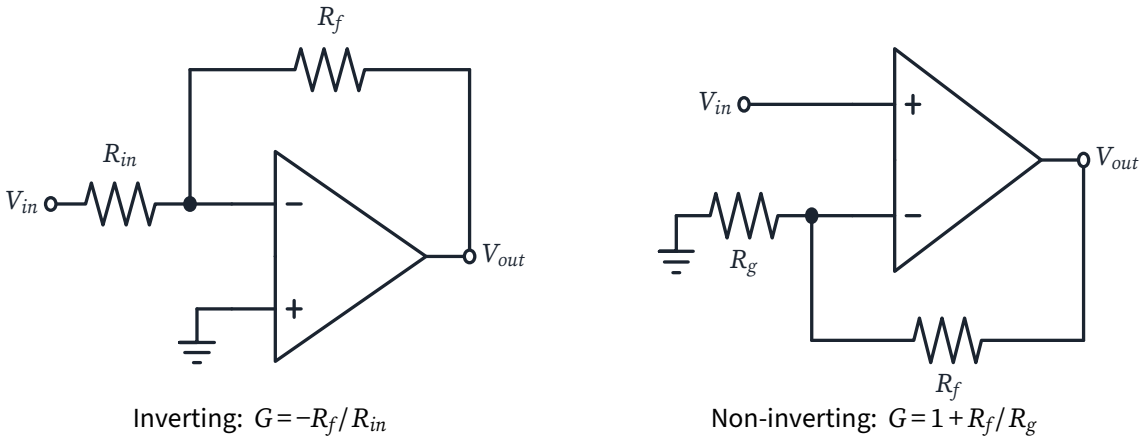


Figure 1.2.2. The two basic configurations. With feedback, the inverting input of the inverting amplifier is a virtual ground; the non-inverting amplifier presents the source with the op-amp's high input impedance.

Filters. A first-order RC low-pass has its −3 dB point at $f_c = 1/(2\pi RC)$ and falls at 20 dB per decade. Two poles fall at 40 dB per decade. The unity-gain Sallen-Key low-pass is the most common second-order active filter: with equal resistors R, the capacitor to the output C_1 and the capacitor to ground C_2,

$$f_0 = \frac{1}{2\pi R\sqrt{C_1 C_2}}, \quad Q = \frac{1}{2}\sqrt{\frac{C_1}{C_2}}.$$

A Butterworth response (maximally flat, $Q = 0.707$) needs $C_1 = 2C_2$. A higher Q peaks and rings, and a lower Q droops early. Phase matters as much as magnitude. A filter delays signals, and in a control loop that delay costs phase margin, which you will account for in Chapter 3.1.

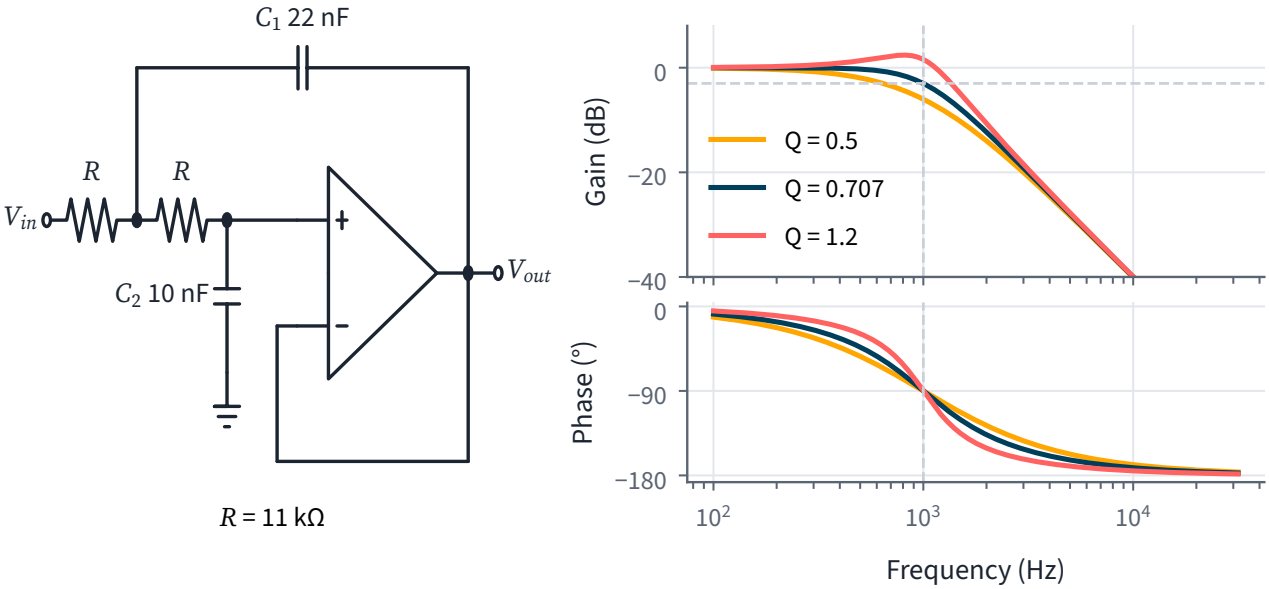


Figure 1.2.3. A 1 kHz Sallen-Key low-pass filter and its response. The measured points are what Lab 1.2.2 should produce; the curves are the second-order model.

Sampling and aliasing. Sampling at f_s cannot distinguish a tone at f from tones at $|f - kf_s|$ for any integer k. Everything above $f_s/2$ (the Nyquist frequency) folds back into the band you care about, indistinguishable from real signal. The only remedy is to remove it before the sampler, either with an analogue anti-alias filter or by sampling so fast that a simple filter suffices and then decimating digitally.

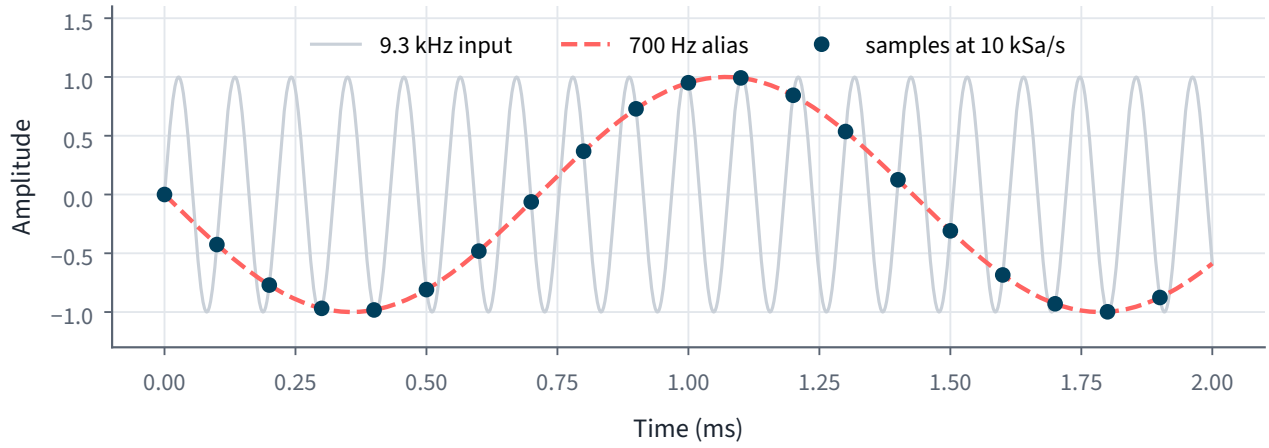


Figure 1.2.4. A 9.3 kHz tone sampled at 10 kSa/s is indistinguishable from a 700 Hz tone. The two give identical samples, so the 9.3 kHz tone can only be removed by an analogue filter in front of the ADC.

Quantisation. An N-bit converter with full-scale range FSR has a step of $\text{LSB} = \text{FSR}/2^N$. Quantisation alone limits the signal-to-noise ratio of a full-scale sine to $6.02N + 1.76$ dB. Real converters add thermal noise, reference noise and nonlinearity, so datasheets quote an effective number of bits, $\text{ENOB} = (\text{SINAD} - 1.76)/6.02$, or noise-free bits. A 24-bit delta-sigma converter with 18 noise-free bits resolves 18 bits, which is still excellent.

Converter architectures. A successive-approximation (SAR) ADC samples instantly onto a capacitor and binary-searches the value. It is fast, has no latency, and needs a driver that can recharge its sampling capacitor in time. A delta-sigma ADC oversamples with a one-bit quantiser and filters digitally. It is slow and precise, and has a built-in sinc filter that rejects some interference (for example 50 and 60 Hz at a chosen data rate) and attenuates, but does not remove, aliases. TI's ADS1115 (16-bit, up to 860 samples per

second) and ADS1220 (24-bit, up to 2,000) are both delta-sigma; the ADC inside the TM4C microcontroller you meet in Unit 2 is a 12-bit SAR.

Noise. A resistor generates thermal noise of density $\sqrt{4kTR}$, about 4 nV/√Hz for 1 kΩ at room temperature; over a bandwidth B the RMS voltage is that density times $\sqrt{B}$. Averaging N independent samples reduces random noise by $\sqrt{N}$, so every factor of four in oversampling buys one bit, until offset, drift and 1/f noise dominate.

References. An ADC measures relative to its reference. A 0.1 % reference makes at best a 0.1 % measurement, whatever the bit count. The remedy for a poor reference is a ratiometric measurement: excite the sensor from the same voltage the ADC uses as its reference, and the reference cancels.

Bridges. A Wheatstone bridge of four strain gauges produces a differential output of a few millivolts per volt of excitation at full load. A 5 kg load cell rated at 2 mV/V gives 10 mV at full scale from 5 V. Measured ratiometrically with a 24-bit ADC and a gain of 128, it has a noise floor of a fraction of a gram.

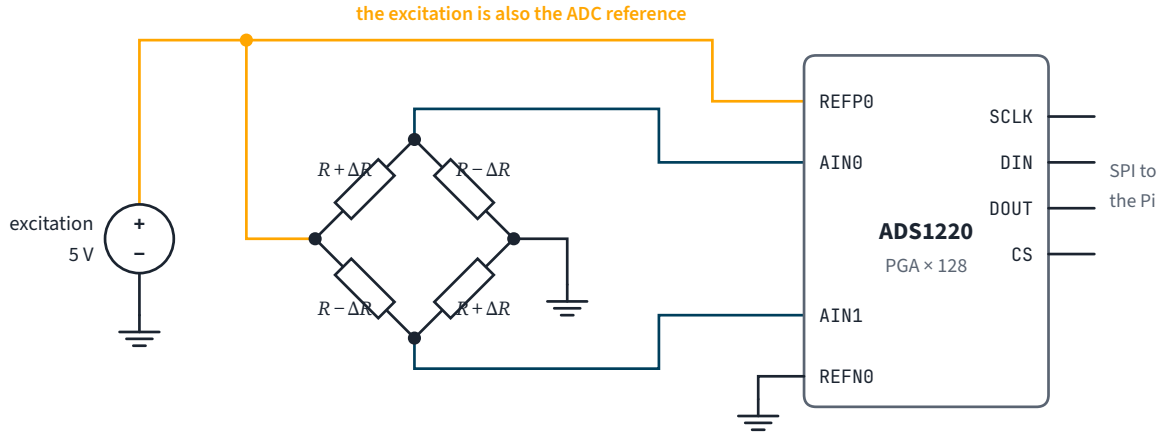


Figure 1.2.5. A load cell measured ratiometrically. The excitation that drives the bridge is also the ADC's reference, so supply drift cancels.

Grounding and interference. A voltage measurement is the difference between the potentials of two wires, so currents flowing in the ground wire add their voltage drop to your signal. Keep sensor returns separate from motor and LED currents, twist signal pairs, and use differential inputs when the sensor is far from the ADC.

Reading

- Horowitz and Hill, *The Art of Electronics*, 3rd ed.: chapter 4 (op-amps: 4.1 to 4.4), chapter 6 (filters: 6.1 to 6.3), chapter 13 sections 13.1 to 13.9 (ADCs and DACs).
- TI, *Op Amps for Everyone* (free; SLOD006): chapters 3, 5 and 16 (Sallen-Key filters).
- The ADS1115 and ADS1220 datasheets, cover to cover: register maps, noise tables, digital filter response.
- Kester (Analog Devices), "MT-001 Taking the Mystery out of the Infamous Formula SNR = 6.02N + 1.76 dB" and "MT-002 What the Nyquist Criterion Means to Your Sampled Data System" (free tutorials).

Labs

LAB 1.2.1 Op-amp stages and their limits — 4 h

Goal. measure GBW, slew rate, offset and output swing, and see where each matters.

Kit. LM358P, TLV2372IP, function generator, oscilloscope, 5 V supply.

1. Build a non-inverting amplifier with $G = 11$ ($R_f = 10$ kΩ, $R_g = 1$ kΩ) on a single 5 V supply, first with the LM358P and then with the TLV2372IP. Drive it with a 100 mV peak-to-peak sine on a 200 mV offset.

2. Measure gain against frequency from 100 Hz to 1 MHz. Find each part's −3 dB point and compare it with GBW divided by 11.
3. Rewire as a unity-gain buffer and drive a 0 to 3 V square wave. Measure the slew rate from the output edge and compare with the datasheet.
4. Ground the input of the $G = 11$ stage and measure the output DC level; compute the input offset voltage.
5. Increase the input until the output clips. Record the highest and lowest output each part reaches on 5 V.

DONE WHEN

- ☐ Measured bandwidths are within a factor of two of GBW/11 for both parts, and the notebook explains the remainder.
- ☐ Slew rates and offsets are measured for both parts and compared with the datasheet's typical and maximum values.
- ☐ A table records the output swing limits and states which part you would use to drive a 0 to 3.3 V ADC input from 5 V.

EVIDENCE FOR THE NOTEBOOK

- Bode-style table and plot per part; slew-rate capture; offset measurement; swing table.

LAB 1.2.2 A second-order filter and its Bode plot — 4 h

Goal. design to a specification, then measure magnitude and phase.

Kit. TLV2372IP, 11 kΩ resistors, 22 nF and 10 nF film or C0G capacitors, function generator, oscilloscope.

1. Design a unity-gain Sallen-Key low-pass with $f_0 = 1$ kHz and $Q \approx 0.707$. Choose preferred values (for example $R = 11$ kΩ, $C_1 = 22$ nF, $C_2 = 10$ nF) and compute the f_0 and Q they give.
2. Build it. At ten frequencies from 100 Hz to 20 kHz measure input and output amplitude and the phase shift (from the time delay between zero crossings).
3. Plot measured magnitude and phase against the model in Python.
4. Drive a 100 Hz square wave and capture the output. Measure the overshoot and relate it to Q.
5. Swap C_1 and C_2 and repeat the square-wave test. Explain what changed.

DONE WHEN

- ☐ The measured −3 dB frequency is within 10 % of your design, and measured phase at f_0 is within 10° of −90°.
- ☐ The plot overlays measurement and model on log axes, committed with the data.
- ☐ The swapped-capacitor result is explained in terms of Q.

EVIDENCE FOR THE NOTEBOOK

- Design calculation; measurement table; overlay plot; two square-wave captures.

LAB 1.2.3 Sampling, aliasing and quantisation with the ADS1115 — 5 h

Goal. characterise a real converter and measure an aliased tone.

Kit. Raspberry Pi 5, ADS1115 breakout, LM4040 2.5 V reference, function generator.

1. Connect the ADS1115 to the Pi's I²C bus. Without a library, write a Python driver from the datasheet's register map: write the configuration register (input multiplexer, gain, data rate, single-shot or continuous) and read the conversion register. Confirm with `i2cdetect` and a logic-analyser capture of one conversion.

2. Short the input (differential, input to input) and record 2,000 samples at each of three gains and three data rates. Compute RMS noise in microvolts and noise-free bits, and compare with the datasheet's noise table.
3. Measure the LM4040's output with the ADS1115 and with the multimeter. Compute both errors against the reference's nominal value and tolerance.
4. At 860 samples per second, sample a 20 Hz sine and reconstruct it. Then sample a 1,000 Hz sine: find the 140 Hz alias in an FFT of your samples, measure its amplitude, and compare with the attenuation predicted by the converter's sinc filter at 1,000 Hz.
5. Average groups of 4, 16 and 64 samples of the shorted input and plot the noise against group size.

DONE WHEN

- ☐ The driver reads correct values at every gain without any library beyond `smbus2`.
- ☐ Your noise table agrees with the datasheet's within a factor of 1.5 at every setting.
- ☐ The alias appears at the predicted frequency, with an amplitude within 3 dB of the sinc-filter prediction.
- ☐ The averaging plot shows noise falling as $1/\sqrt{N}$ and the notebook says where it stops falling and why.

EVIDENCE FOR THE NOTEBOOK

- Driver source; noise table; FFT plot with the alias marked; averaging plot.

LAB 1.2.4 A load cell on a 24-bit converter — 5 h

Goal. build the measurement that becomes Unit 3's thrust stand.

Kit. SparkFun 5 kg straight-bar load cell (TAL220B) with two plywood mounting blocks, Olimex BB-ADS1220 breakout (solder its headers), Raspberry Pi 5, calibration masses (coins, or kitchen weights).

1. Mount the load cell between two plates so that the load passes through it as the maker intends. Wire excitation, signal and the ADS1220's reference inputs ratiometrically as in the figure above.
2. Write an SPI driver for the ADS1220 in Python from its register map: SPI mode 1, gain 128, 20 samples per second with simultaneous 50/60 Hz rejection, external reference on REFP0/REFN0, and the DRDY pin on a Pi GPIO so you read each conversion as it completes.
3. Calibrate with at least five known masses spanning the range (UK coins have specified masses: a £1 coin is 8.75 g and a 2p coin 7.12 g). Fit offset and scale; report the linearity error as a percentage of full scale.
4. Leave 1 kg on the cell for 15 minutes and log the reading every second. Measure the creep.
5. Halve the excitation voltage (if your breakout allows it) and show the calibrated reading barely changes.

DONE WHEN

- ☐ A fitted calibration converts counts to grams with a residual under 0.1 % of full scale at every point.
- ☐ The noise with no load is quoted in grams RMS, and the creep over 15 minutes is quantified.
- ☐ The ratiometric claim is demonstrated, or the notebook explains why your breakout could not test it.

EVIDENCE FOR THE NOTEBOOK

- Wiring photo; driver source; calibration plot with residuals; creep log plot.

Problem set

1. An op-amp with 3 MHz GBW and 0.5 V/µs slew rate amplifies a 20 kHz signal with a gain of 51. What is the closed-loop bandwidth, and what is the largest undistorted output amplitude at 20 kHz?

2. Design a two-pole Butterworth low-pass at 200 Hz in front of an ADC sampling at 1 kSa/s. How much does it attenuate a tone at 800 Hz, which aliases to 200 Hz? What order would you need for 60 dB?
3. Compute the thermal noise of a 100 kΩ source resistance over 1 kHz, and say whether it matters to an ADS1220 at a gain of 128.
4. An ideal 12-bit ADC has a 3.3 V range. What is the LSB, the ideal SNR, and the SNR if the input sine is 10 dB below full scale?
5. A delta-sigma ADC at 10 samples per second has a sinc response with nulls at multiples of 10 Hz. Explain why it rejects both 50 and 60 Hz mains pickup, and what it does to a 25 Hz signal.
6. A 5 kg load cell gives 2 mV/V and is excited at 5 V; the ADS1220 at a gain of 128 has 100 nV RMS of noise at 20 samples per second. What is the resolution in grams? What does averaging ten readings give, at what cost?
7. A sensor 2 m from the ADC shares a ground wire with a fan drawing 300 mA, and the wire has 0.1 Ω of resistance. What error appears on a single-ended reading, and how do you remove it?
8. Show that a ratiometric measurement cancels the excitation voltage, and name the two errors it does not cancel.

Deliverables and stretch

Deliverables. The ADS1115 and ADS1220 drivers in the labs repository, with a README stating their measured noise. The filter design and Bode overlay. The calibrated load-cell rig, labelled and kept, because it becomes the thrust stand of Chapter 3.1.

Stretch. Build a two-stage instrumentation amplifier from the TLV2372IP and compare it with the ADS1220's internal PGA. Measure the ADS1220's internal temperature sensor against the TMP117 you meet in Chapter 1.3.

COMPLETION CHECKLIST

- ☐ **Lab 1.2.1:** bandwidth, slew rate, offset and swing measured for both op-amps and compared with the datasheet.
- ☐ **Lab 1.2.2:** filter within 10 % of design, Bode overlay committed, the Q experiment explained.
- ☐ **Lab 1.2.3:** library-free ADS1115 driver; noise table within a factor of 1.5 of the datasheet; alias found at the predicted frequency and amplitude.
- ☐ **Lab 1.2.4:** load cell calibrated to 0.1 % of full scale with creep quantified; the rig kept for Unit 3.
- ☐ **Problem set:** all eight answered with units; problems 2 and 6 checked numerically in Python.
- ☐ **You can explain**, without notes: GBW and slew rate, why aliasing cannot be fixed after sampling, what ENOB means, and why ratiometric measurement cancels the reference.

CHAPTER 1.3

Sensors with TI silicon

TIME	2.5 weeks
YOU BUILD	Library-free drivers for five TI sensors; a temperature and humidity metrology rig checked against a salt reference; a calibrated power monitor; a magnetic angle sensor; an optical time-of-flight rangefinder; a written error budget for the drone's battery-current measurement.
NEEDS	Raspberry Pi 5, logic analyser, Qwiic cables, the TMP117, humidity, INA228 and TMAG5273 breakouts, the Pololu OPT3101 module, the multimeter's K-type thermocouple.

WHAT THIS BUYS OMH

A baseboard management controller is mostly a sensor hub: inlet and component temperatures, rail voltages, currents and power, fan speeds, intrusion and position switches. A drone is the same hub with an IMU attached. Every one of these readings arrives as a register on a bus, with an accuracy the datasheet states under conditions your board may not meet. This chapter teaches you to read a sensor datasheet critically, write the driver from the register map, and say how wrong the number on the screen can be.

SKIP TEST

1. A temperature sensor quotes ±0.1 °C accuracy and 7.8 m°C resolution. Explain the difference and give one way the reading can still be 1 °C wrong on your board.
2. Write the I^2C transaction (start, address, R/W, register pointer, repeated start, data, ACK and NACK) that reads a 16-bit register from a device at address 0x48.
3. Compute the shunt calibration value for an INA228 with a 1 mΩ shunt and a maximum current of 40 A.
4. Why can a 3D Hall-effect position sensor not be used as a drone's compass?
5. An optical time-of-flight sensor modulates its light at 10 MHz and measures a phase shift of 0.6 rad. How far away is the target, and what is the largest distance it can report without ambiguity?

If all five are easy, do Labs 1.3.3 and 1.3.6 only.

Core ideas

A sensor is a transfer function plus errors. The datasheet gives the ideal relationship between the physical quantity and the output code, then the deviations: offset, gain error, nonlinearity, hysteresis, noise, drift with temperature and time, and response time. **Resolution** is the smallest change the output can show; **precision** (repeatability) is how much repeated readings scatter; **accuracy** is how close the reading is to the truth. A 16-bit sensor can be precise to a millikelvin and inaccurate by a whole kelvin. Datasheets give "typical" values (half of parts are worse) and "maximum" values (guaranteed, usually over a stated range of temperature and supply). Design to the maximums.

Errors combine. Independent random errors add as the root of the sum of squares; correlated and systematic errors add linearly. An error budget lists every term with its source and its magnitude in the measurement's units, then totals them both ways. Spend money only on reducing the largest term.

Calibration removes repeatable errors. A one-point calibration removes offset, two points remove offset and gain, and more points expose nonlinearity. Calibration cannot remove noise, hysteresis or drift that has not happened yet, and no calibration can be more accurate than its reference.

The bus. Most of the sensors in this course use I²C: an open-drain clock and data line with pull-ups (Chapter 1.1), 7-bit addresses, a write of a register pointer followed by a repeated start and a read. Devices may stretch the clock, NACK a bad address, or hang the bus mid-byte if a master resets halfway through a transaction. A driver should detect a stuck data line and clock it free. SPI is faster and simpler: chip select, clock, and data in each direction, in one of four clock modes. On Linux, `/dev/i2c-1` and `/dev/spidev0.0` expose the buses to userspace, and `i2cdetect`, `i2ctransfer` and `spi-pipe` let you talk to a part before you write any code.

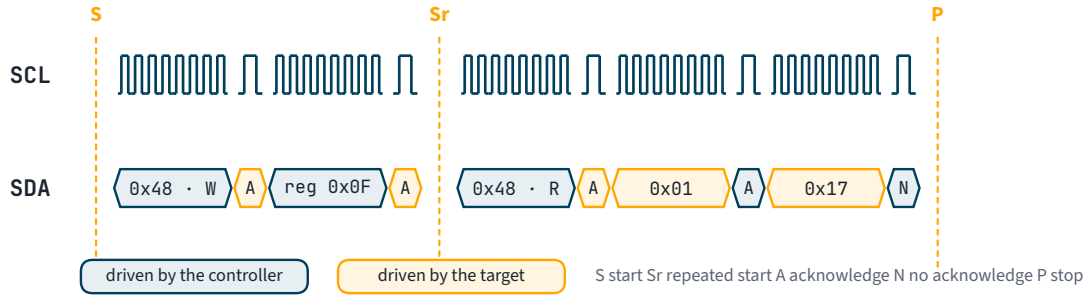


Figure 1.3.1. Reading a 16-bit register over I²C: write the register pointer, repeated start, read two bytes, NACK the last. Lab 1.3.1 asks you to find every one of these features in a logic-analyser capture.

Temperature: silicon bandgap sensors. A transistor's base-emitter voltage falls by about 2 mV per kelvin, and the difference between two transistors at different current densities is proportional to absolute temperature. TI's TMP117 digitises that difference to 7.8125 m°C per LSB and is trimmed and tested to ±0.1 °C from −20 to +50 °C. The sensor measures the temperature of its own die. On your board the die is coupled both to the air you want to measure and to the copper it is soldered to, so placement, airflow and the sensor's own dissipation (self-heating) set the real accuracy. A first-order thermal model, a time constant τ and an offset from self-heating, describes most of what you will see.

Humidity: capacitive polymer sensors. A thin polymer absorbs water and changes the capacitance of an interdigitated electrode. TI's HDC3022 pairs that with a temperature sensor behind a PTFE filter, because relative humidity is meaningless without the temperature it was measured at. Its errors are offset after exposure to high humidity or contaminants, slow response, and condensation. An on-chip heater drives off moisture. Saturated salt solutions give laboratory-grade references at almost no cost. Sodium chloride holds the air above it at about 75 % RH over a wide temperature range.

Light: photodiodes with a shaped response. TI's OPT4048 filters four photodiodes to the CIE X, Y and Z colour-matching responses and a wide channel, auto-ranges over many decades, and reports each channel as an exponent and mantissa. Its errors are angular response, infrared leakage and the window you put in front of it. Linux has no driver for it, so Chapter 4.3 has you write one.

Magnetic: the Hall effect. A current through a semiconductor in a magnetic field develops a transverse voltage proportional to the field. TI's TMAG5273 measures all three axes over ±40 or ±80 millitesla (the A1 variant) and computes the angle of a magnet in hardware. Earth's field is 25 to 65 microtesla. The A1's resolution of 1.2 µT per count is fine enough to register it, but its noise (22 µT RMS on X and Y even with 32× averaging) and its offset (±300 µT typical) are larger than the field itself, so a Hall sensor is used to measure magnets and positions. Compasses use magnetoresistive parts, which Chapter 3.1 adds to the drone.

Current and power: shunts and Kelvin connections. A current through a small resistor develops a small voltage; a precise differential ADC measures it. TI's INA228 measures the shunt voltage (±163.84 mV or ±40.96 mV range) with a 20-bit delta-sigma converter, measures the bus voltage up to 85 V, multiplies them, and accumulates energy and charge in 40-bit registers. Its accuracy is limited by the shunt more often than by the chip: the shunt's tolerance, its temperature coefficient, the voltage drop in the copper between the shunt and the sense pins (which Kelvin, or four-wire, connections remove), and its self-heating at high current. The power the shunt wastes, I^2R, sets an upper limit on its resistance, and the INA228's offset sets a lower limit.

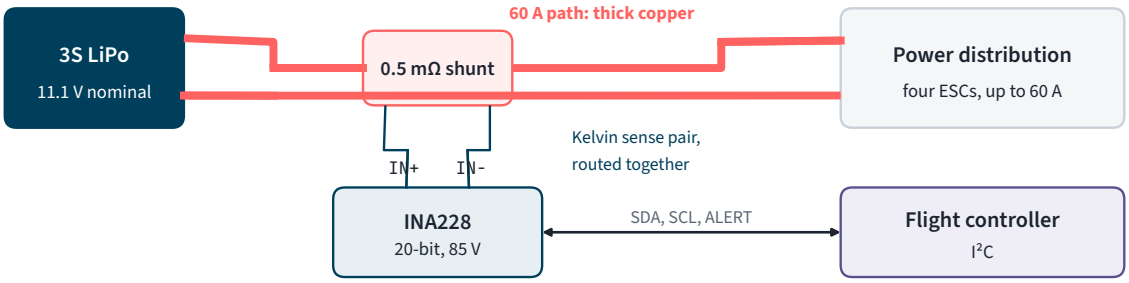


Figure 1.3.2. Measuring battery current with an INA228. The sense lines leave the shunt's pads directly (Kelvin connection) so the copper carrying 60 A does not add its drop to the 30 mV you are measuring.

Distance: time of flight. Distance is half the round trip times the propagation speed. Ultrasonic sensors such as TI's PGA460 time an acoustic echo, so their reading depends on the speed of sound, which varies by about 0.6 m/s per kelvin. Optical sensors such as TI's OPT3101 modulate an infrared LED and measure the phase φ of the returned light: $d = c\varphi/(4\pi f_{\mathrm{mod}})$, unambiguous up to $c/(2f_{\mathrm{mod}})$. Their errors come from ambient light, the target's reflectivity, multipath and crosstalk inside the module, which is why they need calibration. Millimetre-wave radar (Chapter 3.3) measures the same round trip as a frequency.

Inertial sensors, previewed. MEMS accelerometers and gyroscopes are the drone's most important sensors, and TI does not make them, so Chapter 3.1 uses ST's LSM6DSOX. Everything in this chapter applies to them: noise density, bias and its temperature drift, scale-factor error, and calibration.

Reading

- The datasheets for TMP117, HDC3022 (or the SHT45 if you bought it), INA228, TMAG5273 and OPT3101. Read each one completely before its lab: the electrical characteristics table, the typical performance graphs, the register map and the application section.
- NXP, UM10204 *I²C-bus specification and user manual* (free), sections 3 (timing and electrical) and 3.1.16 (bus clear).
- Fraden, *Handbook of Modern Sensors*, 5th ed., chapters 2 (transfer functions), 3 (characteristics) and the chapters for temperature, humidity, magnetic and distance sensing.
- One TI application note on shunt-based current-sensing error analysis (search ti.com for "current sense amplifier error analysis") and the INA228 datasheet's application section on shunt selection.
- Linux `Documentation/i2c/dev-interface.rst` for the userspace I²C interface.

Labs

All labs run on the Raspberry Pi 5 in Python 3 with `smbus2` and `spidev` only. Do not use a sensor library from the vendor or from Adafruit. Write each driver from the register map, and only afterwards compare it with someone else's.

LAB 1.3.1 Bus bring-up and a driver from the datasheet — 4 h

Goal. capture a register read on the bus and write a correct driver.

Kit. Pi 5, TMP117 breakout, Qwiic cables, logic analyser.

1. Enable I²C on the Pi, connect the TMP117, and find it with `i2cdetect -y 1`. Read the device ID register (0x0F) with `i2ctransfer` and confirm it reads 0x0117.
2. Capture the transaction with the logic analyser and annotate start, address, ACK, register pointer, repeated start, data and NACK.

3. Write `tmp117.py`: read temperature (two's complement, 7.8125 m°C per LSB), set conversion cycle and averaging, start one-shot conversions, and set the high and low alert limits.
4. Wire the ALERT pin to a Pi GPIO and use `gpiomon` (or `gpiod` in Python) to log an event when you warm the sensor between your fingers past the high limit.
5. Unplug the sensor in the middle of a read loop and make your driver report a clear error and recover when it is reconnected.

DONE WHEN

- [ ] The annotated capture shows every element of the transaction in the figure above.
- [ ] Your driver's readings match `i2ctransfer` decoded by hand for at least three temperatures.
- [ ] The alert reaches the GPIO and is logged with a timestamp.
- [ ] Unplugging and reconnecting does not crash the program or hang the bus.

EVIDENCE FOR THE NOTEBOOK

- Annotated capture; driver source with docstrings citing datasheet sections; log of the alert event.

LAB 1.3.2 Temperature and humidity metrology — 6 h

Goal. measure accuracy, self-heating, response time and a humidity reference.

Kit. TMP117 and humidity breakouts, the multimeter's thermocouple, a sealed jar or food container, table salt, a small dish.

1. Write a driver for your humidity sensor (`hdc3022.py`, or `sht45.py` for the alternative), including the CRC-8 check on every word the device returns. Reject readings whose CRC fails and count them.
2. Place the TMP117, the humidity sensor and the thermocouple bead together in still air for 30 minutes. Log all three every 10 s and compute the differences and their scatter.
3. Self-heating: run the TMP117 in continuous conversion with no averaging, then in one-shot mode once a minute. Compare the steady readings.
4. Response time: move the sensor pair from room air into a warm airflow (a desk fan blowing over a warm, not hot, surface) and back. Fit a first-order response to each step and report τ for each sensor.
5. Humidity reference: make a slurry of salt and a little water in a dish inside a sealed container with the humidity sensor. Log for 12 hours. The equilibrium should be about 75 % RH. Report the offset and how long equilibrium took.
6. Run the sensor's heater for 10 s and log the effect on both its readings.

DONE WHEN

- [ ] The three temperature readings agree within the sum of their stated accuracies, or the notebook explains the disagreement.
- [ ] Self-heating is quantified in millikelvin and the two time constants are fitted from data.
- [ ] The salt test gives the humidity sensor's offset at 75 % RH with the equilibrium curve plotted.

EVIDENCE FOR THE NOTEBOOK

- Logs as CSV; plots of the drift, step responses and salt test; the fitted parameters.

LAB 1.3.3 Power metrology with the INA228 — 4 h

Goal. measure current, voltage, power and energy, and know the error.

Kit. INA228 breakout, the 12 V fan, bench supply, multimeter, power resistors.

1. Write `ina228.py`: compute CURRENT_LSB and SHUNT_CAL for the breakout's shunt and a 2 A maximum, configure conversion time and averaging, and read shunt voltage, bus voltage, current, power, energy and charge.
2. Put the fan and then a 5 Ω power resistor in series with the shunt. Compare the INA228's current and voltage with the multimeter at five operating points up to 2 A.
3. Accumulate energy for ten minutes while the fan runs at three PWM duties (reuse Lab 1.1.3's switch). Compare with power × time from your own sampling.
4. Configure the shunt-overcurrent alert and show it firing on the Pi GPIO when you stall the fan by hand (gently, with a gloved finger on the hub).
5. Measure the shunt's actual resistance with the four-wire method (a known current from the bench supply and the multimeter across the shunt pads) and recompute your calibration.

DONE WHEN

- ☐ Current agrees with the multimeter within 1 % at every point above 100 mA, after calibrating with the measured shunt value.
- ☐ The energy accumulator and your integrated power agree within 2 %.
- ☐ The overcurrent alert fires within one conversion period of the stall.

EVIDENCE FOR THE NOTEBOOK

- Calibration arithmetic; comparison table; energy plot; alert log.

LAB 1.3.4 Magnetic angle with the TMAG5273 — 3 h

Goal. measure an angle without contact and show why the sensor cannot be a compass.

Kit. TMAG5273 breakout, a diametrically magnetised disc magnet, a protractor or printed angle scale, the 12 V fan or a hobby servo.

1. Write `tmag5273.py`: enable X, Y and Z, set the range, read the three axes in millitesla and the on-chip angle result.
2. Fix the magnet to a shaft above the sensor. Rotate it in 15° steps against the printed scale. Plot the sensor's angle against the true angle and fit the error.
3. Move the magnet 2 mm further away and repeat. Explain the change in field magnitude and in angle error.
4. Remove all magnets and log the field for a minute while you rotate the board. Compute the noise and compare it with Earth's field.

DONE WHEN

- ☐ The angle error is under 2° RMS at the nominal spacing.
- ☐ The notebook shows numerically why the sensor cannot be a compass.

EVIDENCE FOR THE NOTEBOOK

- Angle-error plot at two spacings; noise log.

LAB 1.3.5 Optical time of flight with the OPT3101 — 5 h

Goal. measure distance by phase and find out what calibration corrects.

Kit. Pololu OPT3101 three-channel module (headers not fitted; solder them), tape measure, a white card and a black card as targets, a desk lamp.

1. Wire the module to the Pi's I^2C bus at 3.3 V and find it at 0x58. Read the OPT3101 datasheet's register map and its initialisation and calibration sections. Pololu's Arduino library shows one working start-up sequence. Port it to Python, and cite the datasheet section behind every register you write.

2. Read distance, amplitude and ambient level from the centre channel against the white card at ten distances from 10 cm to 1 m. Plot error against distance.
3. Repeat with the black card, then with the lamp shining at the sensor. Explain the changes in amplitude, noise and error.
4. Apply a phase-offset correction from one measured distance and repeat the sweep. Report what the correction fixed and what it did not.

DONE WHEN

- [ ] Distance error after your one-point correction is under 3 cm from 10 cm to 1 m on the white card.
- [ ] The effects of target reflectivity and ambient light are quantified and explained.

EVIDENCE FOR THE NOTEBOOK

- Driver source with a datasheet citation per register; error plots before and after correction; the three-condition comparison.

LAB 1.3.6 An error budget for the drone's battery current

3 h, paper and Python

Goal. quantify a real measurement before building it.

The drone in Chapter 3.5 draws about 12 A in hover and up to 60 A at full throttle from a three-cell pack, through a 0.5 mΩ shunt measured by an INA228.

1. List every error term: shunt tolerance and temperature coefficient (assume the shunt warms by 40 K at full current), INA228 offset and gain error from the datasheet's maximums, ADC noise at your chosen conversion time and averaging, copper drop if the Kelvin connection is imperfect, and the effect of a 1 kHz ripple on the reading.
2. Express each in milliamps at 1 A, 12 A and 60 A. Total them both as root sum of squares and linearly.
3. Decide the shunt power rating, the ADC range setting and the averaging you would use, and say which single term dominates at 1 A and at 60 A.
4. Verify the low-current end on the bench: the breakout's own shunt at 100 mA and 1 A.

DONE WHEN

- [ ] The budget table has every term with its source and the two totals at three currents.
- [ ] The design choices are justified by the dominant terms.

EVIDENCE FOR THE NOTEBOOK

- The budget as a table and as a Python script; the bench check.

Problem set

1. The TMP117 is trimmed to ±0.1 °C maximum from −20 to +50 °C. List four reasons your reading of the air in a rack module could still be 2 °C wrong, and the design choice that addresses each.
2. Derive the RMS noise in °C of the average of 64 TMP117 conversions if one conversion has 7.8 m°C RMS noise, and say why the real improvement is less.
3. Write the bus-clear procedure for an I^2C slave holding SDA low, and explain why it works.
4. Compute SHUNT_CAL for an INA228 with a 0.5 mΩ shunt, a 60 A maximum and the ±40.96 mV range. What is the current LSB? What full-scale current can the range represent before it saturates?
5. A 1 mΩ shunt carries 30 A. What power does it dissipate, and what is its temperature rise in a 20 K/W mounting? With a temperature coefficient of 50 ppm/K, what error does that cause?
6. The speed of sound in air is about $331.3 + 0.606T$ m/s with T in °C. An uncorrected ultrasonic altimeter calibrated at 20 °C reads 10.00 m on a winter morning at −5 °C. What is the true altitude? Why does an optical time-of-flight sensor not have this error, and what error does it have instead?

7. A Hall sensor has 0.2 mT RMS noise. How many readings must be averaged to resolve Earth's horizontal field of 20 µT with a signal-to-noise ratio of 10, and why is that still not a usable compass?
8. Your humidity sensor reads 79 % above saturated sodium chloride at 25 °C. Write the one-point correction and state what it cannot correct.

Deliverables and stretch

Deliverables. A `sensors/` package in the labs repository with five drivers, each with a docstring per function citing the datasheet section it implements and a test that runs against a recorded bus trace. The metrology logs and plots. The error budget as a document and a script.

Stretch. Port `tmp117.py` to C on the Pi using the `i2c-dev` ioctl interface directly, and measure the time per read against Python. Write a driver for the OPT4048 colour sensor (Adafruit 6335) and convert its XYZ output to a correlated colour temperature; Chapter 4.3 turns it into a kernel driver.

COMPLETION CHECKLIST

- ☐ **Lab 1.3.1:** annotated I^2C capture; TMP117 driver verified against hand-decoded reads; alert on GPIO; hot-unplug recovery.
- ☐ **Lab 1.3.2:** three-way temperature comparison, self-heating, two fitted time constants and the 12-hour salt test.
- ☐ **Lab 1.3.3:** INA228 within 1 % of the multimeter after shunt calibration; energy within 2 %; overcurrent alert demonstrated.
- ☐ **Lab 1.3.4:** angle error under 2° RMS; compass limit shown with numbers.
- ☐ **Lab 1.3.5:** time-of-flight error under 3 cm after correction; reflectivity and ambient light quantified.
- ☐ **Lab 1.3.6:** error budget at three currents with design choices justified.
- ☐ **Problem set:** all eight answered with units.
- ☐ **You can explain**, without notes: accuracy versus precision versus resolution, how to combine errors, why Kelvin connections matter, and how phase becomes distance in a time-of-flight sensor.

CHAPTER 1.4

Power conversion and batteries

TIME	2.5 weeks
YOU BUILD	Measured efficiency, ripple and transient response of a TI buck converter; demonstrated inrush control, short-circuit protection and reverse-polarity protection; a characterised three-cell lithium-polymer pack; two power trees with chosen TI parts that become Board P and Board A in Unit 3.
NEEDS	Bench supply, oscilloscope, DC electronic load, INA228 breakout, LM317 in TO-220, the TI buck, eFuse and ideal-diode EVMs, the drone's 3S 5000 mAh LiPo with its balance charger, LiPo bag and cell checker, the multimeter's thermocouple.

WHAT THIS BUYS OMH

OMH's modules take a 12 V bus and turn it into a dozen rails, one of which feeds a CPU at 0.9 V and 100 A. The drone takes a battery whose voltage falls by a quarter over a flight and sags by more under load. In both, the power system determines efficiency, heat, noise, what happens when something is plugged in live or wired backwards, and how long the system runs. This chapter turns Chapter 1.1's intuition into converters, protection and batteries you have measured.

SKIP TEST

1. A synchronous buck converts 12 V to 3.3 V at 2 A, switching at 500 kHz with a 4.7 µH inductor. What is the duty cycle and the inductor's peak-to-peak ripple current, and what output capacitance keeps the ripple under 10 mV ignoring ESR?
2. Why does a buck converter's efficiency fall at light load, and what do "PFM" and "skip mode" do about it?
3. A 470 µF bulk capacitor is hot-plugged onto a 12 V bus. Estimate the inrush current with 50 mΩ of total path resistance, and say how an eFuse with a dV/dt control limits it.
4. Why does an ideal-diode controller with an N-channel MOSFET waste less power than a Schottky diode at 10 A?
5. A 3S 5000 mAh pack reads 12.4 V at rest and 11.5 V at 30 A. What is its internal resistance, and how much of a 15-minute hover's energy is lost inside it at a steady 12 A?

If all five are easy, do Labs 1.4.4 and 1.4.5 only.

Core ideas

Start from the power tree. Draw the source, each conversion stage, each rail with its voltage, maximum current, tolerance and noise requirement, and each load. Sum the currents upward through the efficiencies. The tree tells you the input current, the heat in each converter, which rails need sequencing, and where you need telemetry. Unit 3 designs two boards from the trees you draw in Lab 1.4.5.

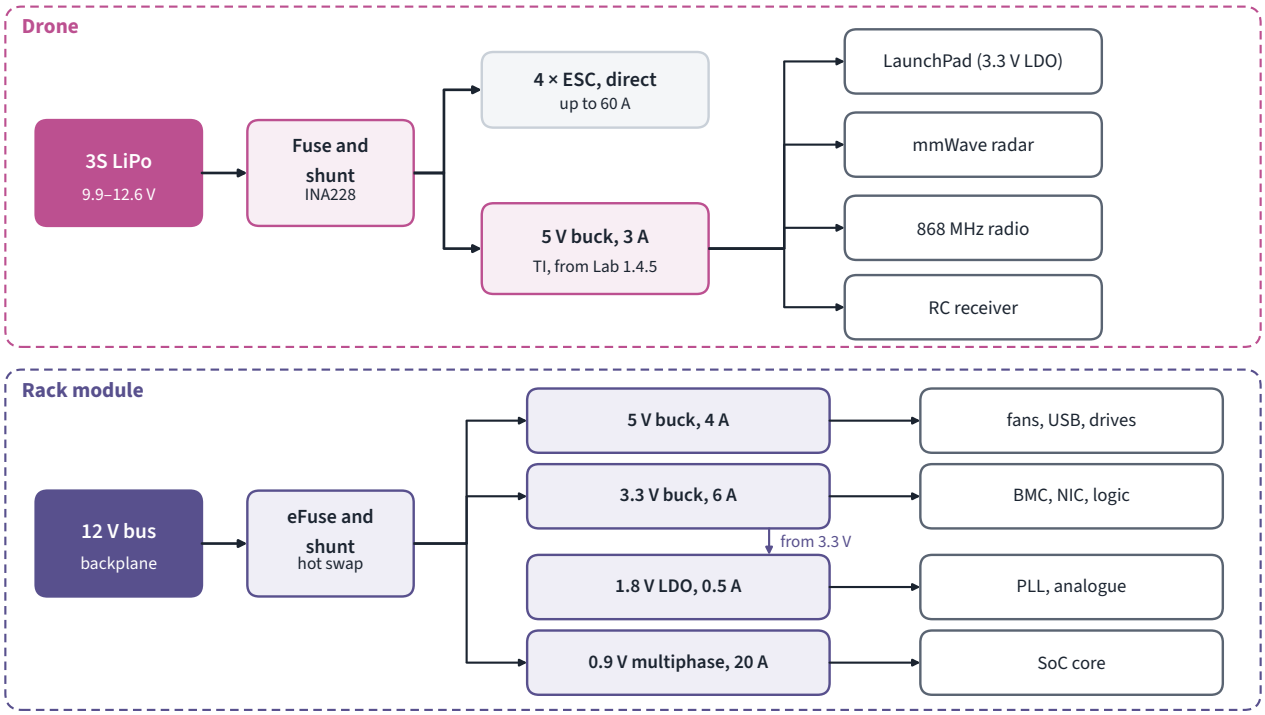


Figure 1.4.1. The two power trees this course builds toward: the drone's (a three-cell pack feeding the ESCs directly and a 5 V converter for the electronics) and a rack module's (a 12 V bus feeding point-of-load converters). Both have an input protection stage and current telemetry.

Linear regulators. A linear regulator is a transistor whose resistance a feedback loop adjusts to hold the output. The dropped voltage times the load current is heat, so efficiency is at best $V_{\text{out}}/V_{\text{in}}$. Linear regulators are quiet, simple and fast. They reject input ripple (PSRR, falling with frequency), and a low-dropout (LDO) part works with as little as 100 to 300 mV of headroom. They suit small currents and noise-sensitive rails such as PLLs, references and ADCs, and they are the second stage after a switcher wherever noise matters.

The buck converter. Two switches alternately connect an inductor to the input and to ground, and the inductor and output capacitor average the result. In continuous conduction the duty cycle is $D \approx V_{\text{out}}/V_{\text{in}}$, and the inductor's ripple current and the output ripple are

$$\Delta I_L = \frac{(V_{\text{in}} - V_{\text{out}})D}{L f_{\text{sw}}}, \quad \Delta V_{\text{out}} \approx \frac{\Delta I_L}{8 f_{\text{sw}} C_{\text{out}}} + \Delta I_L \cdot \text{ESR}.$$

Designers choose L for a ripple of 20 to 40 % of the full-load current. Losses come from conduction (I^2R in the switches and the inductor's winding resistance), switching (the overlap of voltage and current at each edge, proportional to f_{sw}), gate drive ($Q_g V_{\text{GS}} f_{\text{sw}}$) and quiescent current. At light load the fixed losses dominate, so modern controllers drop into pulse-frequency or skip modes, trading ripple for efficiency.

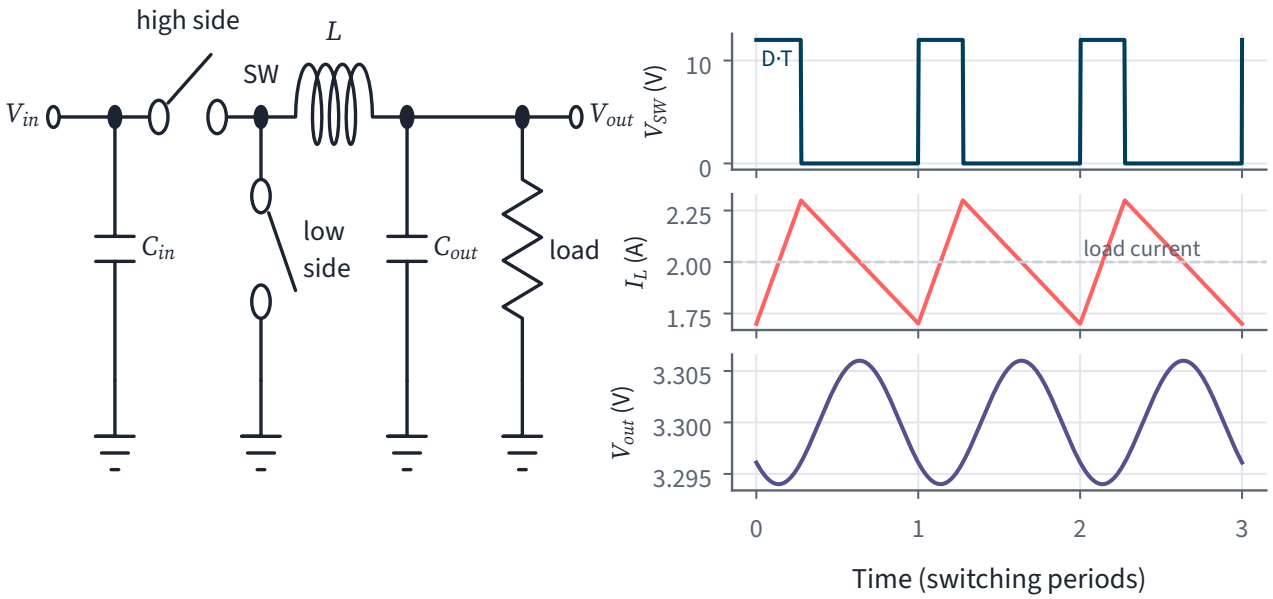


Figure 1.4.2. A synchronous buck converter and its waveforms. The switch node swings between V_{in} and ground at the duty cycle. The inductor current ramps up and down around the load current, and the output ripple is the voltage that this ripple current develops across the output capacitor.

Control and stability. A feedback loop compares the divided output with a reference and sets the duty cycle. Peak-current-mode control, the most common, senses the inductor current each cycle, which makes the inductor behave like a current source and simplifies compensation. The loop's crossover frequency (typically a tenth of f_{sw} or less) and phase margin (45° to 60°) determine how the output recovers from a load step. With too little phase margin the output rings, and with too little bandwidth it sags. Most TI converters are internally compensated for a range of output capacitance, and stepping outside that range is the commonest way to make a stable design oscillate.

The layout adds inductance and radiating loops. The loop from the input capacitor through the high-side switch and back through the low-side switch carries current that changes in nanoseconds. Its area radiates like an antenna and its inductance produces voltage spikes, so keep it as small as possible. Keep the switch node small (it is the noisiest copper on the board) and the feedback trace away from it. Every TI converter datasheet has a layout example, and you should copy it exactly on your first board.

Boost and buck-boost. A boost puts the inductor on the input side and raises the voltage. A buck-boost can do either, which a single-cell battery system needs when the battery voltage falls through the output voltage. TI's BQ25798 is a buck-boost battery charger for one to four cells.

Protection. Faults you must design for:

- **Reverse polarity.** A series diode wastes $V_F \times I$. A P-channel MOSFET in the supply path, or an N-channel MOSFET in the return path, wastes less. An ideal-diode controller such as TI's LM74700 drives an N-channel MOSFET so that it conducts like a diode with millivolts of drop and turns off in microseconds when current reverses.
- **Inrush and hot plug.** Charging bulk capacitance through milliohms draws hundreds of amps for microseconds, so connectors arc and the upstream bus dips. A hot-swap controller or an eFuse such as TI's TPS25947 ramps the output at a controlled $\mathrm{d}V/\mathrm{d}t$, limits current, and latches off or retries on a fault. Every OMH module's 12 V input needs one, because modules are inserted into live racks.
- **Overvoltage and transients.** A transient-voltage suppressor (TVS) diode clamps short spikes; an overvoltage cut-off disconnects sustained ones.
- **Undervoltage.** Lockout (UVLO) keeps a converter off until its input is valid; for a battery, a low-voltage cut-off protects the cells.

Lithium-polymer cells. A lithium-ion cell's open-circuit voltage runs from about 3.0 V empty to 4.20 V full, nominally 3.7 V. Capacity in ampere-hours times nominal voltage is energy in watt-hours; the C-rate expresses current relative to capacity (1C from a 3 Ah cell is 3 A). Each cell has an internal resistance of a few milliohms to tens of milliohms that rises with age and cold, so voltage sags under load and the pack

warms. The voltage-against-charge curve is flat through the middle, which makes voltage a poor fuel gauge under load. Packs put cells in series (3S is three cells, 11.1 V nominal) and sometimes in parallel; series cells drift apart and must be balanced while charging.

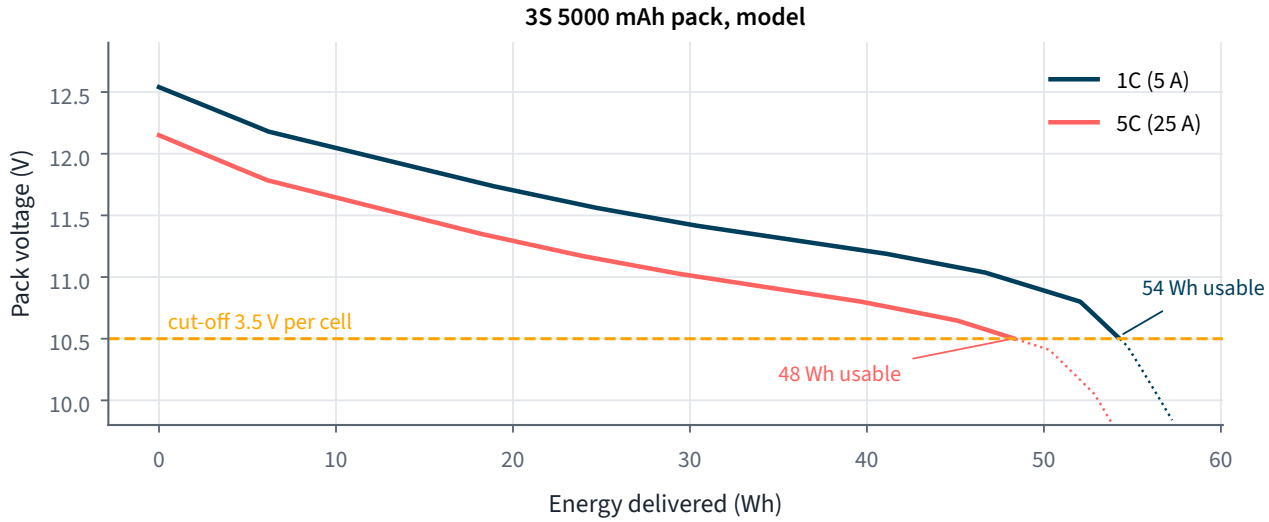


Figure 1.4.3. A model of a three-cell pack discharging at 1C and 5C. Internal resistance shifts the whole curve down, and the usable energy before the 3.5 V per cell cut-off shrinks at high current.

Gauging and protection. Coulomb counting integrates current (the INA228's charge accumulator does it in hardware) but drifts without an anchor. The voltage measured at rest provides that anchor. TI's fuel gauges combine both with a model of the cell's impedance. A battery management system watches every cell for over- and under-voltage, over-current and temperature, and opens a switch when any limit is crossed. A hobby drone pack has none of this, so your flight controller must do the minimum (voltage cut-off and current logging) itself.

Handling. Charge at 1C or less, constant current then constant voltage to no more than 4.20 V per cell, always with balancing, never hot, never below about 5 °C and never unattended. Do not discharge below about 3.0 V per cell; set cut-offs at 3.3 to 3.5 V per cell. Store at 3.6 to 3.8 V per cell in a fire-resistant container. Retire any pack that is swollen, hot or crash-damaged: isolate it, discharge it through a resistive load, never short it, and take it to a battery recycling point.

Reading

- Horowitz and Hill, *The Art of Electronics*, 3rd ed., chapter 9 (voltage regulation and power conversion), sections 9.1 to 9.7.
- Erickson and Maksimović, *Fundamentals of Power Electronics*, 3rd ed., chapters 1 to 3 (steady-state converter analysis) and chapter 9 sections 9.1 to 9.4 (control loops), as deep as you want to go.
- The datasheets and EVM user's guides for your buck converter, the TPS25947 and the LM74700, including their layout sections.
- TI, *Power Supply Design Seminar* papers on buck converter fundamentals and on loop compensation (free on ti.com; search "power supply design seminar").
- Battery University (batteryuniversity.com), articles BU-205 (lithium types), BU-409 (charging lithium-ion), BU-501a (discharge characteristics) and BU-802 (capacity loss).

Labs

LAB 1.4.1 Linear versus switching — 3 h

Goal. measure where the power is lost in a linear and a switching regulator.

Kit. LM317 in TO-220 with its two resistors, the TI buck EVM, bench supply, DC load, thermocouple meter.

1. Set an LM317 for 5.0 V from 12 V. Load it to 300 mA with the DC load. Measure input and output power, the regulator's case temperature after five minutes with no heatsink (it dissipates about 2 W and gets hot enough to burn), and its dropout as you lower the input.
2. Configure the TI buck EVM for 5 V (or use its fixed output) from 12 V and repeat at 300 mA and at 1 A.
3. Compute each efficiency and compare the measured temperature rise with $P\theta_{JA}$ from the datasheets.
4. Scope both outputs with the spring ground tip and record ripple and noise at the same scale.

DONE WHEN

- ☐ An efficiency table at both currents agrees with the textbook expectation (V_{out}/V_{in} for the linear) within 3 points.
- ☐ Measured temperature rises are within 30 % of the θ_{JA} prediction, or the discrepancy is explained.

EVIDENCE FOR THE NOTEBOOK

- Efficiency table; thermocouple readings; two ripple captures.

LAB 1.4.2 Inside a buck converter — 6 h

Goal. measure duty cycle, ripple, efficiency and transient response against the equations.

Kit. TI buck EVM, bench supply, DC load, INA228, the MOSFET load-step switch from Lab 1.1.3 with a power resistor.

1. Probe the switch node with a short ground spring at 12 V in and half load. Measure frequency and duty cycle. Sweep the input from 8 to 16 V and plot duty cycle against V_{out}/V_{in}.
2. From the switch-node waveform and the inductor value in the EVM's bill of materials, compute ΔI_L. Measure the output ripple and compare with the formula, with and without an extra 22 µF ceramic.
3. Measure efficiency from 10 mA to the EVM's maximum current in at least eight steps, using the bench supply's readings for input power and the INA228 for output power. Mark where the converter changes mode.
4. Build a load step from 10 % to 90 % of full load with the MOSFET switch and a power resistor, driven at 100 Hz. Capture the output's undershoot, overshoot and settling time.
5. Read the EVM's layout and label the hot loop on a photograph.

DONE WHEN

- ☐ Duty cycle tracks V_{out}/V_{in} within 5 % and the mode change at light load is identified on the efficiency plot.
- ☐ Computed and measured output ripple agree within a factor of two, with the remaining difference explained (ESR, ESL, probing).
- ☐ The transient capture gives undershoot in millivolts and settling time in microseconds, compared with the datasheet's load-transient figure.

EVIDENCE FOR THE NOTEBOOK

- Duty-cycle plot; ripple captures; efficiency curve; transient capture; annotated EVM photo.

LAB 1.4.3 Protection: inrush, short circuit and reverse polarity — 4 h

Goal. capture inrush, short-circuit and reverse-polarity events with and without protection.

Kit. TPS25947 eFuse EVM, LM74700 ideal-diode EVM, a 1,000 µF low-ESR capacitor, a 0.1 Ω current-sense resistor, a 1N5819 Schottky diode.

1. Without the eFuse, hot-plug the 1,000 µF capacitor (through the 0.1 Ω sense resistor) onto the 12 V supply set to a 3 A limit. Capture the inrush current from the voltage across the sense resistor.

2. Repeat through the eFuse EVM. Set its current limit and dV/dt capacitor per the user's guide for a 20 ms ramp. Capture the ramp and the current.
3. Short the eFuse output through a 0.1 Ω resistor and capture the response. Note whether it latches or retries and how long it takes to act.
4. Through the LM74700 EVM, reverse the input polarity (current-limited) and show no reverse current flows. Then measure its forward drop at 2 A and compare with the 1N5819's, keeping the diode's 2 A measurement brief because it is rated for 1 A.

DONE WHEN

- ☐ Captures show the inrush peak without protection, the controlled ramp with it, and the short-circuit response with its timing.
- ☐ The reverse-polarity test is captured and the forward-drop comparison is expressed in watts at 2 A and extrapolated to 20 A.

EVIDENCE FOR THE NOTEBOOK

- Four captures with current scales; the power comparison.

LAB 1.4.4 Characterise the drone battery

5 h plus charging

Goal. measure the energy the drone can draw from its pack.

Kit. Overlander 5000 mAh 3S pack, SkyRC S65 charger, LiPo bag, cell checker, Siglent SDL1020X-E electronic load, INA228 with the Bourns 1 mΩ shunt, TMP117, multimeter.

> **SAFETY**
> Charge and discharge inside the LiPo bag on a non-flammable surface, attended, with the charger's balance lead connected. Stop at 3.5 V per cell. If a cell swells or warms abnormally, stop and follow the handling rules above.

1. Charge the pack fully with balancing. Measure each cell's resting voltage at the balance connector.
2. Measure internal resistance: apply 2 A, then step to 10 A for two seconds, and compute $\Delta V / \Delta I$ for the pack and, from the balance connector, for each cell.
3. Discharge at a constant 0.5C until any cell reaches 3.5 V, logging pack voltage, current, energy and pack temperature (TMP117 taped to the pack) every second. Recharge, then repeat at 2C.
4. Plot both discharge curves against delivered watt-hours. Compute the usable energy to the cut-off at each rate.
5. Return the pack to storage voltage (3.8 V per cell) and record it.

DONE WHEN

- ☐ Pack and per-cell internal resistances are measured and the weakest cell identified.
- ☐ Two discharge curves are plotted with usable watt-hours and temperature rise at each rate.
- ☐ The notebook states the energy you will budget for flight, with its margin.

EVIDENCE FOR THE NOTEBOOK

- Cell table; discharge logs and plots; storage-voltage record.

LAB 1.4.5 Two power trees

5 h, paper and TI's parametric search

Goal. specify the power systems you will build in Unit 3.

1. **Drone.** From the 3S pack: the four ESCs directly; 5 V at 3 A for the flight-controller LaunchPad, the radar, the telemetry radio and the receiver; 3.3 V for sensors. Add input protection, a current shunt

with the INA228, and a battery-voltage divider for an ADC. Choose a TI buck for the 5 V rail, justify it with an efficiency estimate at 1 A, and draw the tree.

2. **Rack module.** From a 12 V bus: an eFuse at the input; 5 V at 4 A; 3.3 V at 6 A; a 1.8 V LDO at 0.5 A; and a 0.9 V core rail at 20 A. Choose TI parts for each stage, estimate losses at 85 % and 92 % converter efficiency, and total the heat.
3. For each tree state the power-up sequence, the telemetry points, and what fault each protection stage handles.

DONE WHEN

- [ ] Both trees are drawn with part numbers, currents, efficiencies and losses at every node.
- [ ] Each choice cites the datasheet figure (efficiency curve, current limit, dropout) that justifies it.

EVIDENCE FOR THE NOTEBOOK

- The two trees as diagrams in the repository, with a parts table.

Problem set

1. Derive the buck's duty cycle from volt-second balance on the inductor. Show why it is independent of load in continuous conduction and what changes in discontinuous conduction.
2. A buck converts 12 V to 1.0 V at 20 A at 400 kHz. Compute the duty cycle and explain why such a low duty cycle is hard. How do multiphase converters help?
3. An LDO with 60 dB of PSRR at 1 kHz and 20 dB at 1 MHz follows a buck with 20 mV of ripple at 1 MHz. What ripple reaches the output? What would you add?
4. Size the dV/dt ramp of an eFuse so that charging 2,200 µF draws no more than 1 A, and compute the energy the eFuse dissipates during the ramp from 0 to 12 V.
5. A three-cell pack with 6 mΩ per cell feeds four motors drawing 3 A each in hover and 15 A each at full throttle. Compute the voltage sag, the power lost in the pack, and the efficiency of delivery to the ESCs in both cases.
6. Explain why a pack's resting voltage after a flight overstates its charge if measured immediately, and how a gauge corrects for it.
7. Your drone's 5 V rail must ride through a 200 ms dip of the battery to 10 V when the motors spin up. Is a buck from the battery enough, or does the rail need a hold-up capacitor? Size it.
8. A rack module's 12 V eFuse trips at 20 A. The module draws 14 A in steady state and 26 A for 5 ms while its drives spin up. Propose two designs that avoid nuisance trips.

Deliverables and stretch

Deliverables. The buck converter measurements as a report with plots; the protection captures; the battery characterisation with the flight-energy budget; the two power trees, which Chapter 3.4 turns into Board P and Board A.

Stretch. Measure the buck's loop gain by injecting a signal across a 10 Ω resistor in the feedback path through an injection transformer driven by the function generator, and estimate crossover and phase margin. Read a TI fuel-gauge datasheet (for example BQ34Z100-R2) and design how the drone would use it.

COMPLETION CHECKLIST

- [ ] **Lab 1.4.1:** efficiency and temperature table for the LM317 and the buck at two currents.
- [ ] **Lab 1.4.2:** duty-cycle sweep, ripple comparison, efficiency curve with the mode change marked, and a load-step capture.

- [ ] **Lab 1.4.3:** inrush with and without the eFuse, short-circuit response, reverse polarity, and the ideal-diode power comparison.
- [ ] **Lab 1.4.4:** per-cell resistance, two discharge curves, usable energy, and the pack left at storage voltage.
- [ ] **Lab 1.4.5:** both power trees with parts, losses, sequencing and telemetry points, committed.
- [ ] **Problem set:** all eight answered with units.
- [ ] **You can explain**, without notes: volt-second balance, where a buck's losses come from, why hot-plug needs an eFuse, and why a LiPo's voltage is a poor gauge under load.

CHAPTER 1.5

Signals on wires and RF fundamentals

TIME	2.5 weeks
YOU BUILD	A time-domain reflectometry measurement of a cable; a calibrated vector network analyser and a set of measured S-parameters; a quarter-wave antenna tuned to 868 MHz; a protected spectrum-analyser setup; an FM receiver written from scratch in Python on an RTL-SDR; link budgets for the drone's radios.
NEEDS	Oscilloscope, SN74HC14N, a 10 m BNC coaxial cable with tee and 50 Ω terminator, tinySA Ultra, LiteVNA 64, RTL-SDR Blog V4, Mini-Circuits 10, 20 and 30 dB SMA attenuators, SMA cables, adapters and terminators, two Linx 868 MHz whip antennas, an SMA bulkhead socket and a scrap of copper-clad board.

WHAT THIS BUYS OMH

Above some frequency every wire must be treated as a transmission line, and every trace radiates. For OMH that frequency arrives with PCIe, DDR and 25 Gbit Ethernet, where impedance, reflections and loss determine whether a board works. For the drone it arrives with the 868 MHz telemetry link, the 2.4 GHz control link and the 60 GHz radar. Signal integrity, EMC and radio rest on the same theory. This chapter teaches its language (decibels, impedance, reflection, S-parameters, noise) and its instruments, so that you can design and test Unit 3's radios and boards by calculation and measurement.

SKIP TEST

1. Convert +20 dBm into milliwatts and into volts RMS across 50 Ω. A signal passes through 30 dB of attenuation and a cable with 1.5 dB of loss: what reaches the far end?
2. A 50 Ω line is terminated in 100 Ω. What are the reflection coefficient, the VSWR and the return loss? What fraction of the power reaches the load?
3. At what length does a PCB trace carrying an edge with a 1 ns rise time start to behave as a transmission line, assuming 6 ps/mm of delay?
4. Compute the free-space path loss at 868 MHz over 2 km. With +14 dBm transmitted, 2 dBi antennas at each end and a receiver sensitivity of −105 dBm, what is the fade margin?
5. What is the thermal noise floor in a 100 kHz bandwidth, and what does a 6 dB noise figure do to it?

If all five are easy, do Labs 1.5.3 and 1.5.5 only.

Core ideas

Decibels. A ratio of powers in decibels is $10\log_{10}(P_2/P_1)$, and a ratio of voltages across the same impedance is $20\log_{10}(V_2/V_1)$. Absolute power is quoted against a reference: dBm against 1 mW, so 0 dBm is 1 mW, +20 dBm is 100 mW and +30 dBm is 1 W. Gains and losses in a chain add in decibels. Antenna gain is quoted in dBi (against an ideal isotropic radiator) or dBd (against a half-wave dipole, which has 2.15 dBi). Noise and interference add as powers, so convert to milliwatts before adding them.

Phasors and impedance. A sine at one frequency is a rotating vector. A resistor, capacitor and inductor relate voltage and current by a complex impedance: R, $1/(j\omega C)$ and $j\omega L$. Series LC circuits resonate at $f_0 = 1/\left(2\pi\sqrt{LC}\right)$, where the reactances cancel. The quality factor Q is 2π times the ratio of the energy stored to the energy dissipated per cycle, and it sets the bandwidth.

Transmission lines. A pair of conductors with distributed inductance L and capacitance C per unit length carries waves at $v = 1/\sqrt{LC}$, which is $c/\sqrt{\varepsilon_{\text{eff}}}$: about 0.66 c in solid-polyethylene coax and about 6 ps/mm on FR-4 microstrip. The ratio of voltage to current in a travelling wave is the characteristic impedance $Z_0 = \sqrt{L/C}$, which is set by geometry and does not depend on length. When a wave meets a load Z_L, a fraction reflects:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}, \quad \text{VSWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|}, \quad \text{RL} = -20 \log_{10}|\Gamma|.$$

An open end reflects the whole wave with the same sign, a short reflects it inverted, and a matched load absorbs it. A wire must be treated as a transmission line, rather than a node, once its one-way delay is a significant fraction of the signal's rise time. A common rule is one sixth. With 1 ns edges that is about 25 mm of FR-4, regardless of the clock frequency.

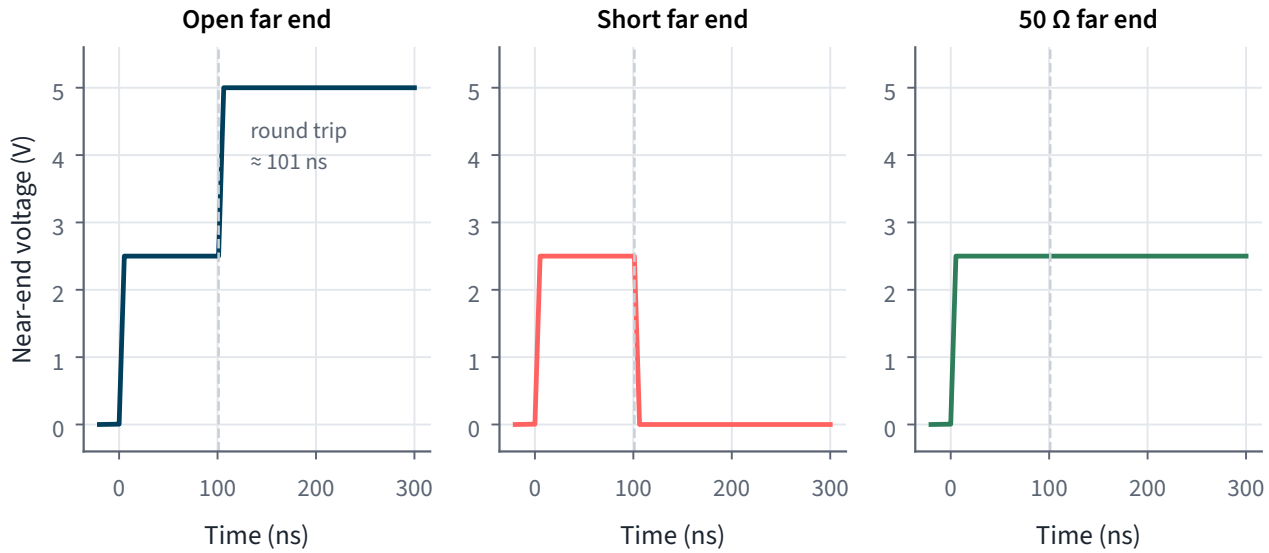


Figure 1.5.1. A step launched into 10 m of coax with three terminations, as Lab 1.5.1 will capture it. The time to the reflection gives the cable's velocity; its sign and size give the termination.

The Smith chart. Every passive impedance maps to a point inside the unit circle of the reflection coefficient Γ. Circles of constant resistance and arcs of constant reactance are drawn on it, so a series inductor moves a point along a resistance circle and a length of line rotates it around the centre. Matching networks are designed on the chart, and a network analyser uses it to display an antenna's impedance.

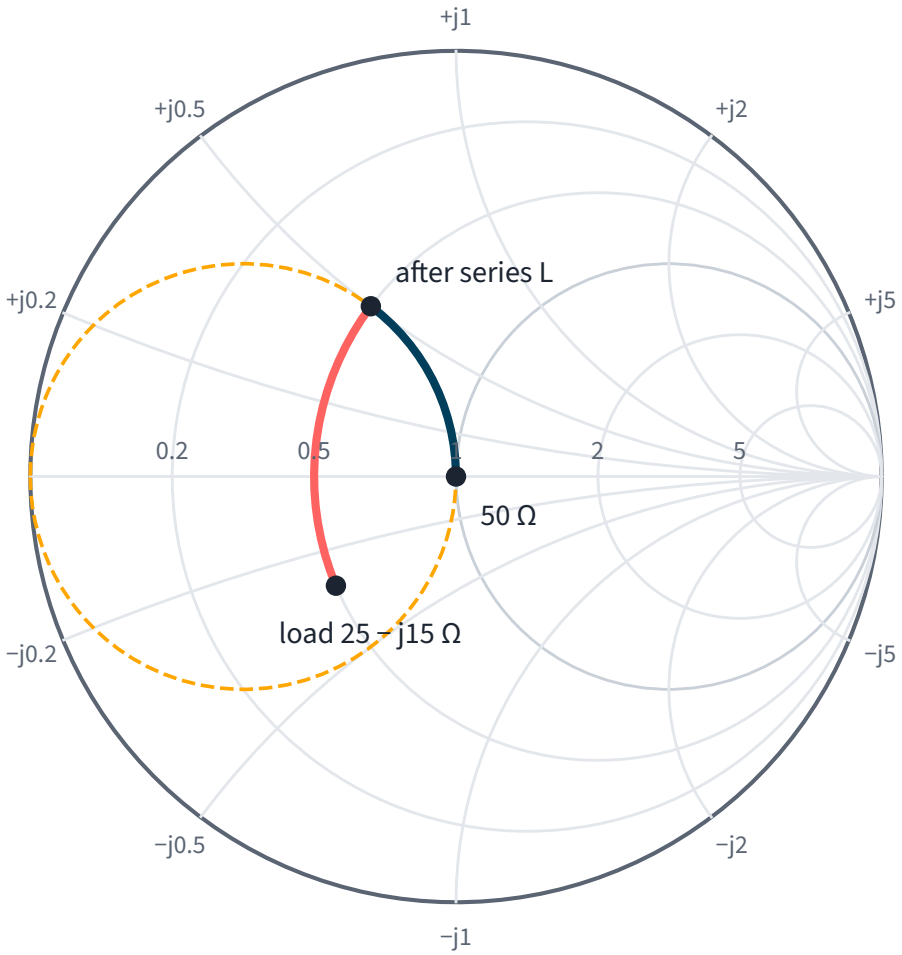


Figure 1.5.2. Matching a 25 − j15 Ω load to 50 Ω at 868 MHz with an L-network: a series inductor moves the point to the unit-conductance circle, then a shunt capacitor moves it to the centre.

S-parameters. At radio frequencies the voltage and current at a port are hard to measure, and the incident and reflected waves are easy to measure. A two-port is described by four complex numbers per frequency: S_{11} (reflection at the input), S_{21} (transmission forward), S_{12} (transmission backward) and S_{22} (reflection at the output). A cable has S_{21} close to 0 dB minus its loss; a 20 dB attenuator has $S_{21} = -20$ dB and small S_{11}; a filter's S_{21} is its passband and stopband. A vector network analyser measures them after a calibration with known standards (short, open, load and through) removes its own errors and those of its cables. Uncalibrated measurements are meaningless above a few hundred megahertz.

Antennas. An antenna is a matched transition from a transmission line to free space. A half-wave dipole is resonant at about 0.47 to 0.48 free-space wavelengths and has 2.15 dBi of gain; a quarter-wave monopole needs a ground plane to supply the missing half. An antenna's gain adds no power. It comes from concentrating power in some directions at the expense of others. Near metal, carbon fibre or a hand, an antenna detunes and its pattern distorts, which is why antenna placement on a drone must be checked by measurement. Polarisation must match between the two ends, or a few to 20 dB is lost.

Propagation and the link budget. In free space the received power falls with the square of distance and of frequency:

$$\mathrm{FSPL} = 20\log_{10}(d) + 20\log_{10}(f) + 32.44 \quad (d \text{ in km}, f \text{ in MHz}).$$

The link budget adds transmitter power and antenna gains, subtracts path loss and cable losses, and compares the result with the receiver's sensitivity. The difference is the fade margin, which must cover multipath, obstruction of the Fresnel zone (the ellipsoid around the line of sight that must be mostly clear, about 9 m in radius at the midpoint of a 1 km link at 868 MHz) and antenna nulls. A drone flying low loses some of this margin to the ground.

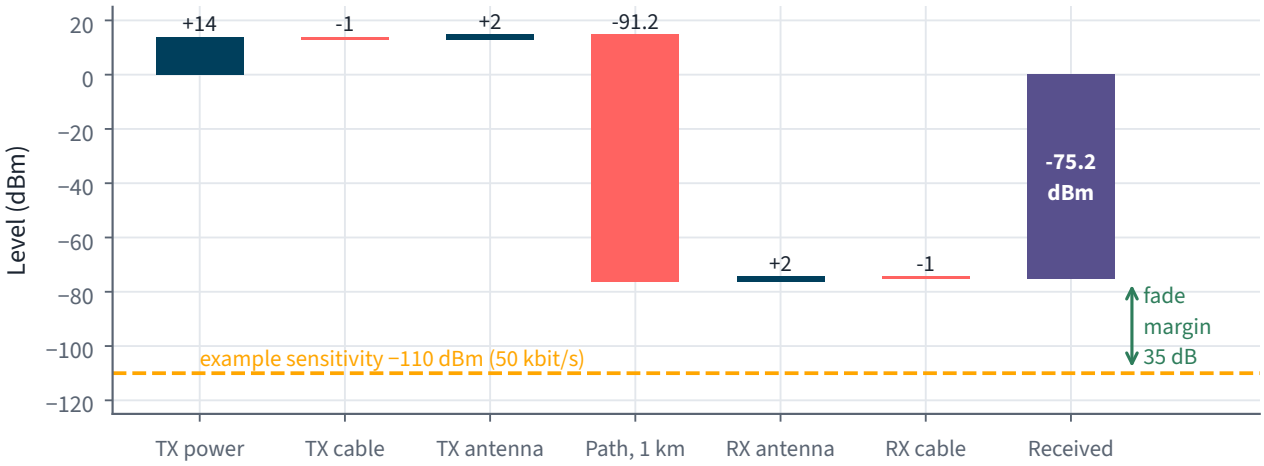


Figure 1.5.3. The 868 MHz telemetry link budget at 1 km, stage by stage. The fade margin is the power left over to cover the losses that the free-space model omits.

Noise. A matched resistor at 290 K delivers kTB of noise power, −174 dBm in each hertz of bandwidth. A receiver's noise figure is how much worse it is than that. Sensitivity is approximately $-174 + 10\log_{10} B + \mathrm{NF} + \mathrm{SNR}_{\min}$ dBm, so halving the data rate halves the bandwidth and buys 3 dB. The sensitivity tables in radio datasheets are this equation evaluated at each data rate.

Instruments and their input limits. A spectrum analyser sweeps a narrow filter (the resolution bandwidth) across a band and shows power against frequency. Narrower RBW lowers the displayed noise floor by $10\log_{10}$ of the ratio and slows the sweep. Its input is a mixer with a maximum safe power of typically a few dBm, so a radio transmitting +20 dBm connected directly will destroy it. Always put enough attenuation in front, and know the arithmetic: +20 dBm through 30 dB is −10 dBm. A software-defined radio digitises a band around a tuned frequency as complex samples, $I + jQ$, and does everything else in software. Because the samples are complex, negative and positive frequencies around the centre are distinct.



Figure 1.5.4. The conducted test setup used from here on. The attenuators protect the instrument, set the received level, and stop the transmitter seeing a mismatch.

Reading

- Bogatin, *Signal and Power Integrity, Simplified*, 3rd ed., chapters 1 to 3 and 7 to 8 (transmission lines and reflections).
- Pozar, *Microwave Engineering*, 4th ed., chapters 2 (transmission-line theory), 4.3 (S-parameters) and 5.1 (L-section matching). Dense; read for the ideas and do the worked examples.
- The ARRL Handbook or Antenna Book chapters on dipoles, monopoles and feed lines, for intuition from people who build antennas.
- Lyons, *Understanding Digital Signal Processing*, 3rd ed., chapter 8 (quadrature signals) before Lab 1.5.5.
- The tinySA Ultra and your VNA's user guides, including their specified maximum input levels.

Labs

SAFETY

Never connect any transmitter output directly to the tinySA, the VNA or the RTL-SDR. Keep at least 40 dB of attenuation between a +20 dBm radio and any of them, and check the arithmetic in the notebook before you connect anything.

LAB 1.5.1 Reflections on a cable — 3 h

Goal. observe reflections on a transmission line in the time domain.

Kit. SN74HC14N oscillator from Lab 1.1.4, 10 m BNC coax, BNC tee, 50 Ω terminator, oscilloscope.

1. Drive the coax through a 47 Ω series resistor from one SN74HC14N output running at about 100 kHz. Probe the near end with the spring ground tip.
2. Capture the near-end waveform with the far end open, shorted and terminated in 50 Ω.
3. From the delay to the reflection compute the cable's velocity factor; compare with the cable's datasheet.
4. Replace the 47 Ω source resistor with a short and explain the extra steps in terms of a second reflection at the source.

DONE WHEN

- [ ] Three captures show the open, short and matched cases with the reflection labelled.
- [ ] The velocity factor is within 5 % of the datasheet value.

EVIDENCE FOR THE NOTEBOOK

- The four captures; the velocity calculation.

LAB 1.5.2 The network analyser and S-parameters — 4 h

Goal. calibrate a network analyser and check its measurements against known parts.

Kit. LiteVNA 64 with its calibration kit, SMA cables and adapters, an Amphenol 50 Ω terminator, the 10, 20 and 30 dB attenuators, the two Linx 868 MHz antennas.

1. Calibrate from 50 MHz to 3 GHz with short, open, load and through at the ends of your test cables. Save the calibration.
2. Measure the terminator (S_{11} should be better than −30 dB), then an open and a short, and find them on the Smith chart.
3. Measure S_{21} of each attenuator and of a cable; compare with their ratings.
4. Measure S_{11} of each 868 MHz antenna in free space. Read off the resonant frequency, the −10 dB bandwidth and the VSWR at 868 MHz.
5. Remove the calibration and repeat one measurement to see how wrong an uncalibrated instrument is.

DONE WHEN

- [ ] Each attenuator measures within 0.5 dB of its rating up to 1 GHz.
- [ ] Both antennas' resonance, bandwidth and VSWR at 868 MHz are tabulated.
- [ ] The uncalibrated comparison is captured.

EVIDENCE FOR THE NOTEBOOK

- Exported Touchstone (.s1p and .s2p) files; screenshots of the Smith chart and S_{21} traces.

LAB 1.5.3 Build and tune an antenna — 3 h

Goal. tune a resonant antenna and measure how nearby materials detune it.

Kit. SMA bulkhead socket, copper-clad board at least 15 cm square, 1 mm copper wire, LiteVNA 64.

1. Compute the free-space quarter-wavelength at 868 MHz. Solder a wire 5 % longer than that to the bulkhead's centre pin, with the bulkhead mounted through the middle of the copper-clad ground plane.
2. Measure S_{11} and trim the wire in 2 mm steps until the resonance is at 868 MHz. Record length against resonant frequency.
3. Bring a hand, then an aluminium plate, then a carbon-fibre sheet (or the drone frame) to 2 cm from the antenna and record the shift.

DONE WHEN

- ☐ The antenna resonates within 10 MHz of 868 MHz with return loss better than 15 dB.
- ☐ The detuning table shows the shift for each nearby material.

EVIDENCE FOR THE NOTEBOOK

- Length-against-frequency table; S_{11} captures before and after trimming; the detuning table.

LAB 1.5.4 Using the spectrum analyser safely — 3 h

Goal. read a spectrum and understand the instrument's settings and limits.

Kit. tinySA Ultra, RTL-SDR Blog V4, attenuators, an antenna for each.

1. With an antenna, find the FM broadcast band, a mobile-network downlink and Wi-Fi at 2.4 GHz. Measure the noise floor at RBW settings of 3 kHz, 30 kHz and 300 kHz and compare the change with $10 \log_{10}$ of the RBW ratio.
2. Use the tinySA's signal-generator output at −30 dBm through 20 dB of attenuation into the RTL-SDR. Predict the level at the SDR and check it. The tinySA Ultra's input is rated +6 dBm absolute maximum and 0 dBm for normal use; keep below both.
3. Build a 10 + 20 dB chain and measure it with the generator and the analyser. Confirm the dB arithmetic.
4. Write the attenuation you will use in Chapter 3.2 for the CC1312R's +14 dBm maximum into each instrument, with the maximum input of each.

DONE WHEN

- ☐ The noise-floor change with RBW agrees with $10 \log_{10}$ of the ratio within 2 dB.
- ☐ The protection plan for each instrument is written with numbers from its manual.

EVIDENCE FOR THE NOTEBOOK

- Screenshots of the three bands and the RBW comparison; the protection plan.

LAB 1.5.5 An FM receiver from scratch — 6 h

Goal. understand complex baseband by building a receiver in software.

Kit. RTL-SDR Blog V4, its antenna, Python with NumPy and SciPy.

1. Capture 10 seconds of complex samples at 2.4 MSa/s centred 250 kHz away from a strong FM station (to keep the station away from the DC spike). Plot the spectrum and a waterfall.
2. Shift the station to 0 Hz by multiplying by a complex exponential, low-pass filter and decimate to about 240 kSa/s.
3. Demodulate with the polar discriminator, the angle of $x[n] \cdot \overline{x[n-1]}$, then apply 50 µs de-emphasis (the European standard) and decimate to 48 kHz. Write a WAV file and listen.
4. Measure the signal-to-noise ratio of the demodulated audio in a quiet passage, and repeat with the antenna connected through 20 dB of attenuation.

DONE WHEN

- [ ] The WAV file is intelligible broadcast audio, produced by your code from raw samples with no SDR library beyond the capture.
- [ ] The spectrum, waterfall and SNR comparison are in the notebook.

EVIDENCE FOR THE NOTEBOOK

- The notebook or script; the plots; a short audio clip.

LAB 1.5.6 Link budgets for the drone

3 h, paper and Python

Goal. decide whether the drone's radios have enough range before you build them.

1. For the 868 MHz telemetry link: transmit power at the level you choose in Chapter 3.2, 2 dBi antennas, 1 dB of cable loss at each end, and the CC1312R's sensitivity from its datasheet at your intended data rate. Compute the free-space range and the range with a 20 dB fade margin.
2. For the 2.4 GHz control link: the receiver's published sensitivity and the transmitter's power. Compute the same two ranges.
3. Plot received power against distance for both links on one log axis with the sensitivity lines.
4. Compute the first Fresnel radius at the midpoint of each link at 500 m and say how high the drone and the ground antenna must be.

DONE WHEN

- [ ] Both budgets are tabulated with sources for every number, and the plot is committed.
- [ ] The notebook states the data rate and power you will use for telemetry and why.

EVIDENCE FOR THE NOTEBOOK

- Budget tables; the plot; the Fresnel calculation.

Problem set

1. A receiver's sensitivity is −110 dBm at 50 kbit/s. Estimate it at 5 kbit/s and at 500 kbit/s, assuming bandwidth scales with data rate. What does the drone gain and lose at each extreme?
2. A 50 Ω coax with velocity factor 0.66 is 3 m long. How long does a reflection take to return? A 2 ns edge is launched; is this cable a transmission line for it?
3. Design an L-network to match 12 Ω to 50 Ω at 868 MHz. Give both topologies and the component values, and say which you would choose for harmonic suppression.
4. A quarter-wave transformer matches 50 Ω to 100 Ω. What impedance and electrical length does it need, and what happens to the match at 1.5 times the design frequency?
5. Two signals at −60 dBm and −63 dBm fall in the same channel. What is the total power?
6. A spectrum analyser's displayed noise floor is −120 dBm at 10 kHz RBW. Can it see a −135 dBm/Hz noise-like signal 20 MHz wide? What setting helps?
7. A radio with a 6 dB noise figure must decode a signal that needs 10 dB SNR in a 150 kHz bandwidth. What is its sensitivity?
8. Explain why an RTL-SDR shows a spike at its centre frequency and why the FM lab tunes 250 kHz away.

Deliverables and stretch

Deliverables. Touchstone files and captures from the VNA; the tuned antenna, kept for Chapter 3.2; the FM receiver script; the drone's link budgets with your chosen telemetry data rate and power.

Stretch. Build an 868 MHz low-pass filter from 0805 C0G capacitors and an air-wound inductor on copper-clad board, design it on the Smith chart and measure it on the VNA. Wind an H-field near-field probe from semi-rigid coax and find the loudest frequencies of the buck EVM from Chapter 1.4 with the tinySA.

COMPLETION CHECKLIST

- ☐ **Lab 1.5.1:** three termination captures and a velocity factor within 5 %.
- ☐ **Lab 1.5.2:** calibrated measurements of terminator, attenuators, cable and antennas, with Touchstone files committed.
- ☐ **Lab 1.5.3:** antenna tuned within 10 MHz of 868 MHz with return loss better than 15 dB; detuning table.
- ☐ **Lab 1.5.4:** RBW and noise-floor comparison; the instrument protection plan written.
- ☐ **Lab 1.5.5:** intelligible FM audio from your own demodulator.
- ☐ **Lab 1.5.6:** both link budgets with sources, the plot, and the chosen telemetry rate and power.
- ☐ **Problem set:** all eight answered.
- ☐ **You can explain**, without notes: decibel arithmetic, the reflection coefficient, what each S-parameter means, the link-budget equation, and why the instruments need attenuators.

UNIT 1

Sign-off

A unit is complete when every chapter's completion checklist is ticked, its evidence is in your notebook and labs repository, and you can do each review task below without notes. Date each line as you finish it. If someone who builds hardware for a living will review your work, ask them to sign the last line.

	Chapter	Checklist ticked	Evidence filed	Date
1.1	Electricity, components and measurement	☐	☐	
1.2	The analogue signal chain	☐	☐	
1.3	Sensors with TI silicon	☐	☐	
1.4	Power conversion and batteries	☐	☐	
1.5	Signals on wires and RF fundamentals	☐	☐	

UNIT REVIEW

Each task takes an hour or less if the unit has done its work. Do them aloud or in writing, from memory, then check against your notebook.

- ☐ At a whiteboard, trace the drone's battery-current measurement from the shunt to a number in software: every component, every term of your Lab 1.3.6 error budget, and which term dominates at hover and at full throttle.
- ☐ Explain from your Lab 1.1.4 captures why a decoupling capacitor belongs next to the pin, and predict what changes if every edge becomes ten times faster.
- ☐ For the ADS1115 and the ADS1220, state the resolution, the measured noise-free bits and the anti-alias filter you would need for a 1 kHz signal, from your own data.
- ☐ Draw the buck converter's switching loop and account for every loss in your measured efficiency curve.
- ☐ Close the 868 MHz link budget at 1 km from memory, and say what the fade margin must cover.

Signed: student _____ Date _____

Reviewed by (optional) _____ Date _____